Adaptive Finite Element Methods for Elliptic PDEs

A Short Course

Lecture Notes (9 Lectures)

Course notes

Contents

1 Basic Theory of PDE

We motivate the variational framework through a model minimization problem, derive its Euler–Lagrange equations, and introduce the function spaces (weak derivatives, Sobolev spaces) in which the theory is well posed.

1.1 The minimization problem

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with $\partial\Omega$ smooth enough to integrate by parts (say C^1 or polygonal). Given a continuous datum $f \in C(\bar{\Omega})$, consider the *energy* (Dirichlet) functional

$$J[v] = \tfrac{1}{2}\int_\Omega |\nabla v|^2 \,\mathrm{d}x - \int_\Omega f\, v \,\mathrm{d}x.$$

We work, for now, in the classical admissible set of functions that are smooth up to the boundary and satisfy the homogeneous Dirichlet condition,

$$\mathcal{A} = \left\{ v \in C^2(\bar{\Omega}) : v = 0 \text{ on } \partial\Omega \right\}, \tag{1}$$

and consider the minimization problem

$$\text{find } u \in \mathcal{A} \text{ such that } \ J[u] = \min_{v\in\mathcal{A}} J[v]. \tag{2}$$

Note $J[v]$ only involves first derivatives, so it is finite for every $v \in \mathcal{A}$ (indeed for every $v \in C^1(\bar{\Omega})$); we ask for C^2 so that the Euler–Lagrange equation below makes classical sense.

1.2 Variational formulation

Suppose $u \in \mathcal{A}$ minimizes J. Fix a *test function* $v \in C_0^2(\bar{\Omega})$ (smooth, vanishing on $\partial\Omega$), so that $u + tv \in \mathcal{A}$ for all $t \in \mathbb{R}$. The scalar function $g(t) = J[u + tv]$ has a minimum at $t = 0$, and

$$g(t) = J[u] + t\Big(\int_\Omega \nabla u \cdot \nabla v - \int_\Omega f v \Big) + \tfrac{t^2}{2} \int_\Omega |\nabla v|^2 .$$

Thus $g'(0) = 0$ gives the *variational (weak) formulation*: find $u \in \mathcal{A}$ such that

$$\mathcal{B}[u, v] := \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x = \int_\Omega f v \,\mathrm{d}x =: \langle f, v\rangle \qquad \forall\, v \in C_0^2(\bar{\Omega}). \tag{3}$$

Conversely, since J is convex (the quadratic term $\frac{t^2}{2}\int_\Omega |\nabla v|^2 \ge 0$), any $u \in \mathcal{A}$ satisfying (3) is a minimizer. So minimization and the variational identity are equivalent on $\mathcal{A}$.

1.3 Euler–Lagrange equations

Since $u \in C^2$, we may integrate (3) by parts; the boundary term drops because $v = 0$ on $\partial\Omega$, giving

$$\int_\Omega (-\Delta u - f)\, v \,\mathrm{d}x = 0 \qquad \forall\, v \in C_0^2(\bar{\Omega}).$$

As $-\Delta u - f$ is continuous and this holds for all such v, the fundamental lemma of the calculus of variations forces the integrand to vanish pointwise. The *strong (Euler–Lagrange) form* is therefore the Poisson problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{4}$$

More generally, for $J[v] = \frac{1}{2}\int_\Omega A\nabla v \cdot \nabla v - \int_\Omega f v$ with a symmetric positive-definite coefficient $A(x)$, the same computation yields $-\operatorname{div}(A\nabla u) = f$.

1.4 Why classical spaces are not enough

We have shown: *if* a classical minimizer $u \in \mathcal{A}$ exists, it solves (4). But does a minimizer exist in $\mathcal{A}$? The direct method of the calculus of variations suggests how to build one: J is bounded below (complete the square and use the Poincaré inequality), so take a *minimizing sequence* $v_n \in \mathcal{A}$ with $J[v_n] \to \inf_{\mathcal{A}} J$. The natural quantity controlled along the sequence is the *energy*

$$\|v\|_{\mathcal{B}[,]} := \Big(\int_\Omega |\nabla v|^2 \, \mathrm{d}x \Big)^{1/2},$$

and one checks (v_n) is Cauchy in this norm. The trouble is that $\big(\mathcal{A}, \ \|\cdot\|_{\mathcal{B}[,]}\big)$ is *not complete*: the limit of a sequence of smooth functions with bounded energy need not be C^2, or even C^1. A Cauchy minimizing sequence may converge to a function with a corner or mild singularity — exactly the behavior we expect near reentrant corners (Lecture 4) — which lies outside $\mathcal{A}$.

This is the same obstruction seen from the operator side. The clean existence tool for a symmetric problem is the following classical result.

Theorem 1.1 (Riesz Representation). *Let H be a Hilbert space with inner product $\langle \cdot, \cdot \rangle_H$ and induced norm $\|\cdot\|_H$, and let $F\colon H \to \mathbb{R}$ be a bounded linear functional. Then there exists a* unique *$u_F \in H$ such that*

$$F(v) = \langle u_F, v \rangle_H \qquad \forall\, v \in H,$$

and $\|u_F\|_H = \|F\|_{H^*} := \sup_{v \neq 0} |F(v)| / \|v\|_H$.

In words: every bounded linear functional on H is *represented* by taking the inner product with a unique fixed element $u_F \in H$. Because $\mathcal{B}[\cdot, \cdot]$ is *symmetric*, it is itself an inner product on any subspace where it is coercive, so the variational problem (3) asks precisely for the Riesz representative of $F(v) = \langle f, v \rangle_{L^2}$ with respect to this inner product. To apply Theorem 1.1 we need a function space that

1. is *complete* for the energy norm (so minimizing/Galerkin sequences have limits in the space), and

2. requires only *one* derivative, in an integral sense, since that is all $\mathcal{B}[u, v]$ and $J[v]$ ever use.

The remedy is to enlarge $\mathcal{A}$ to the *completion* of $C_0^\infty(\bar{\Omega})$ under the energy norm. This completion is the Sobolev space $H_0^1(\Omega)$ introduced below; it is a Hilbert space, and on it the Riesz theorem applies with coercivity supplied by the Poincaré inequality $\|v\|_{L^2} \le C_\Omega \|\nabla v\|_{L^2}$ and continuity by Cauchy–Schwarz. The remainder of this lecture builds the machinery (weak derivatives, Sobolev spaces, embeddings) needed to make this rigorous. Section 1.11 extends the existence theory to nonsymmetric problems, where the Riesz argument no longer applies and one must invoke the Lax–Milgram theorem.

1.5 Weak derivatives

The variational formulation (3) only requires *one* derivative of u and v, and only in an integral sense. We first make precise what "a derivative in an integral sense" means.

Definition 1.2 (weak derivative)**.** Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$ a multi-index with $|\alpha| = \alpha_1 + \cdots + \alpha_d$. We say $g \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak derivative* $D^\alpha u$ if

$$\int_\Omega u \, D^\alpha \varphi \, \mathrm{d}x = (-1)^{|\alpha|} \int_\Omega g \, \varphi \, \mathrm{d}x \qquad \forall\, \varphi \in C_c^\infty(\Omega).$$

The identity is integration by parts with boundary terms absent because φ is compactly supported in Ω. Weak derivatives are unique up to sets of measure zero; when the classical derivative exists and is locally integrable it coincides with the weak one.

Examples. *1. Classical functions.* If $u \in C^k(\Omega)$, the weak derivative $D^\alpha u$ equals the classical derivative for $|\alpha| \le k$.

2. Piecewise smooth functions. Let $\Omega = \Omega_1 \cup \gamma \cup \Omega_2$ where $\gamma = \partial\Omega_1 \cap \partial\Omega_2$ is a smooth interface, and suppose $u_i \in C^1(\overline{\Omega}_i)$ with matching traces $u_1|_\gamma = u_2|_\gamma$. Then the weak gradient is the piecewise classical gradient:

$$\nabla u(x) = \begin{cases} \nabla u_1(x), & x \in \Omega_1, \\ \nabla u_2(x), & x \in \Omega_2. \end{cases} \tag{5}$$

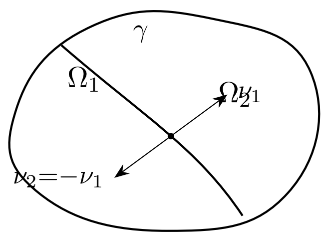


Figure 1: Two subdomains meeting at interface γ with opposite outward normals. Provided traces match, the weak gradient is the piecewise classical gradient.

Proof of (5). Integrate by parts on each subdomain:

$$\begin{aligned} -\int_\Omega u \nabla\varphi \,\mathrm{d}x &= -\int_{\Omega_1} u_1 \nabla\varphi \,\mathrm{d}x - \int_{\Omega_2} u_2 \nabla\varphi \,\mathrm{d}x \\ &= \int_{\Omega_1} \nabla u_1 \,\varphi \,\mathrm{d}x + \int_{\Omega_2} \nabla u_2 \,\varphi \,\mathrm{d}x - \int_\gamma u_1 \varphi \,\nu_1 \,\mathrm{d}s - \int_\gamma u_2 \varphi \,\nu_2 \,\mathrm{d}s. \end{aligned}$$

Since $u_1 = u_2$ on γ and $\nu_2 = -\nu_1$, the interface terms cancel. A key application: *hat functions* ϕ_z (continuous, piecewise linear, equal to 1 at node z and supported on the element star ω_z) satisfy (5), so their weak gradients are the piecewise-constant gradients $\nabla\phi_z|_T$.

3. Radial functions.

Exercise 1.3. Let $\Omega = B(0,1) \subset \mathbb{R}^d$ and $f(x) = \rho(|x|)$ where $\rho\colon (0,1] \to \mathbb{R}$ is smooth with $\int_0^1 |\rho'(r)|\, r^{d-1}\,\mathrm{d}r < \infty$. Show that $\nabla f(x) = \rho'(|x|)\, x/\,|x|$ for $x \ne 0$.

4. The borderline case: $W^1_d \not\hookrightarrow L^\infty$.

Example 1.4. On $\Omega = B(0,1) \subset \mathbb{R}^d$, consider $u(x) = \log\log \dfrac{2}{|x|}$. By the exercise,

$$\nabla u(x) = -\frac{1}{\log\frac{2}{|x|}} \frac{x}{|x|^2}, \qquad \int_\Omega |\nabla u|^d \,\mathrm{d}x = C_d \int_0^1 \frac{1}{\left(\log\frac{2}{r}\right)^d} \frac{\mathrm{d}r}{r} < \infty,$$

so $u \in W^1_d(\Omega)$, yet u is unbounded near 0. Hence $W^1_d(\Omega) \not\hookrightarrow L^\infty(\Omega)$ (the Sobolev number $\mathrm{sob}(W^1_d) = 1 - d/d = 0$ is the borderline where L^∞ fails). By superposition, taking $\sum_i \alpha_i u(\cdot - x_i)$ with x_i dense and $\alpha_i \to 0$ fast, one can construct W^1_d functions that are infinite on a prescribed dense set.

5. *The fundamental solution.*

Example 1.5. The functions

$$\Phi(x) = \begin{cases} -\dfrac{1}{2\pi} \log |x|, & d = 2, \\ \dfrac{1}{4\pi |x|}, & d = 3, \end{cases}$$

satisfy $\langle -\Delta\Phi, \varphi \rangle = \varphi(0)$ for all $\varphi \in C_c^\infty(\mathbb{R}^d)$, i.e. $-\Delta\Phi = \delta_0$ weakly, where δ_0 is the Dirac mass at the origin. Note $\Phi \in W^1_{p,\mathrm{loc}}(\mathbb{R}^d)$ for $p < d/(d-1)$, but $\Phi \notin H^1_{\mathrm{loc}}(\mathbb{R}^d)$ for $d \geq 2$.

1.6 Sobolev spaces and the Sobolev number

Definition 1.6 (Sobolev space)**.** For $1 \leq p \leq \infty$ and $m \in \mathbb{N}_0$,

$$W_p^m(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \ \forall\, |\alpha| \leq m\},$$

with norm and seminorm (for $1 \leq p < \infty$)

$$\|u\|_{W_p^m} = \Big(\sum_{|\alpha| \leq m} \|D^\alpha u\|_{L^p}^p \Big)^{1/p}, \qquad |u|_{W_p^m} = \Big(\sum_{|\alpha| = m} \|D^\alpha u\|_{L^p}^p \Big)^{1/p}.$$

We write $H^m = W_2^m$ and define $H_0^1(\Omega)$ as the closure of $C_c^\infty(\Omega)$ in H^1.

Exercise 1.7. Show that $W_p^m(\Omega)$ is a *Banach space*. *Hint:* embed it isometrically as a closed subspace of $\bigoplus_{|\alpha| \leq m} L^p(\Omega)$ via $u \mapsto (D^\alpha u)_{|\alpha| \leq m}$.

Remark 1.8 ($H^m(\Omega)$ is a Hilbert space)**.** For $p = 2$ the norm arises from the inner product $\langle u, v \rangle_{H^m} = \sum_{|\alpha| \leq m} \int_\Omega D^\alpha u \, D^\alpha v \, \mathrm{d}x$, making $H^m(\Omega)$ a Hilbert space. In particular, $H_0^1(\Omega)$ carries the inner product $\langle u, v \rangle_{H_0^1} = \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x$, which is equivalent to the full H^1 inner product by the Poincaré–Friedrichs inequality (Lemma 1.15, proved in Section 1.9).

Well-posedness of the Poisson problem in $H_0^1(\Omega)$. We now have all the pieces to make the existence argument of Section 1 rigorous. The weak formulation (3) set in $H_0^1(\Omega)$ reads: find $u \in H_0^1(\Omega)$ such that

$$\mathcal{B}[u, v] = \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x = \int_\Omega f \, v \, \mathrm{d}x =: F(v) \qquad \forall\, v \in H_0^1(\Omega). \tag{6}$$

The left-hand side is the H_0^1 inner product $\langle u, v \rangle_{H_0^1} = \mathcal{B}[u, v]$; that $\mathcal{B}[\cdot, \cdot]$ really is an inner product on $H_0^1(\Omega)$ rests on the zero-trace Poincaré–Friedrichs inequality, stated and proved later as Lemma 1.15 in Section 1.9, which forces $\mathcal{B}[v, v] = \|\nabla v\|_{L^2}^2 = 0 \Rightarrow v = 0$. The right-hand side defines a bounded linear functional: by Cauchy–Schwarz and the same Poincaré–Friedrichs inequality $\|v\|_{L^2} \leq C_\Omega \|\nabla v\|_{L^2}$,

$$|F(v)| \leq \|f\|_{L^2} \|v\|_{L^2} \leq C_\Omega \|f\|_{L^2} \|\nabla v\|_{L^2} = C_\Omega \|f\|_{L^2} \|v\|_{H_0^1},$$

so $F \in (H_0^1(\Omega))^*$. The Riesz Representation Theorem 1.1 (with $H = H_0^1(\Omega)$) therefore guarantees a *unique* solution $u \in H_0^1(\Omega)$, satisfying the stability bound

$$\|u\|_{H_0^1(\Omega)} = \|F\|_{(H_0^1)^*} \leq C_\Omega \|f\|_{L^2(\Omega)}.$$

The single most useful bookkeeping device is the *Sobolev number*

$$\operatorname{sob}(W^m_p) = m - \frac{d}{p}. \tag{7}$$

It appears in every scaling argument. Let $\widehat{x} = \lambda x$ be a dilation $\Omega \to \widehat{\Omega}$ and set $\widehat{v}(\widehat{x}) = v(x)$. Then $D^\alpha v(x) = \lambda^{|\alpha|} D^\alpha \widehat{v}(\widehat{x})$, and changing variables ($\mathrm{d}\widehat{x} = \lambda^d \,\mathrm{d}x$),

$$\|D^\alpha v\|_{L^p(\Omega)} = \lambda^{|\alpha|-d/p} \, \|D^\alpha \widehat{v}\|_{L^p(\widehat{\Omega})}.$$

At the top order $|\alpha| = m$ the exponent is exactly $m - d/p = \operatorname{sob}(W^m_p)$: this is the power of λ that $|\cdot|_{W^m_p}$ acquires under dilation. A bounded, scale-invariant embedding $W^m_p \hookrightarrow W^k_q$ is therefore only possible when $\operatorname{sob}(W^m_p) \geq \operatorname{sob}(W^k_q)$.

1.7 Sobolev embeddings

Theorem 1.9 (Sobolev embedding)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain, $m > k \geq 0$, $1 \leq p, q \leq \infty$. Then*

$$W^m_p(\Omega) \hookrightarrow W^k_q(\Omega)$$

provided $\operatorname{sob}(W^m_p) \geq \operatorname{sob}(W^k_q)$, *i.e. $m - d/p \geq k - d/q$, except when equality holds and $q = \infty$. In that case $\|v\|_{W^k_q} \leq C\, \|v\|_{W^m_p}$ for all $v \in W^m_p(\Omega)$, and the embedding is* compact *when the inequality is strict and $m > k$.*

Examples. *1. $W^1_p \hookrightarrow L^q$ (gaining integrability).* Take $m = 1$, $k = 0$: the embedding $W^1_p \hookrightarrow L^q$ holds for $1 - d/p \geq -d/q$, i.e. $q \leq dp/(d-p)$ when $p < d$. For $d = 2$ and $p = 2$: $H^1(\Omega) \hookrightarrow L^q(\Omega)$ for all $1 \leq q < \infty$ (any finite exponent).

2. The borderline sob $= 0$ *fails for L^∞.* For $p = d$ we have $\operatorname{sob}(W^1_d) = 0$. Example 4 above shows $u(x) = \log\log\frac{2}{|x|} \in W^1_d(B_1) \setminus L^\infty(B_1)$, confirming $W^1_d(\Omega) \not\hookrightarrow L^\infty(\Omega)$. (The embedding does hold into L^q for all $q < \infty$.)

3. Morrey embedding into Hölder spaces. If $\operatorname{sob}(W^1_p) = 1 - d/p > 0$ (i.e. $p > d$), then $W^1_p(\Omega) \hookrightarrow C^{0,\alpha}(\overline{\Omega})$ with $\alpha = 1 - d/p$.

4. Why Lipschitz? Embeddings can fail on non-Lipschitz domains. Consider the cusp $\Omega = \{(x,y) \in \mathbb{R}^2 : |y| < x^\gamma,\ 0 < x < 1\}$, $\gamma > 1$, which is not Lipschitz. For $p > 2$ and small $\varepsilon > 0$ with $p < \gamma + 1 - \varepsilon$, the function $v(x,y) = x^{-\varepsilon/p}$ satisfies $v \in W^1_p(\Omega)$ yet $v \notin L^\infty(\Omega)$, so the embedding $W^1_p \hookrightarrow L^\infty$ fails.

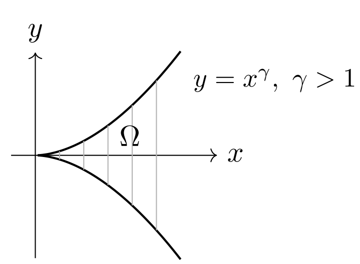


Figure 2: A cusp domain ($\gamma > 1$): not Lipschitz. The Sobolev embedding $W^1_p \hookrightarrow L^\infty$ ($p > 2$) fails on this domain.

Remark 1.10 (Lipschitz functions). On a Lipschitz domain, $W^1_\infty(\Omega) = C^{0,1}(\overline{\Omega})$: the two conditions "$v$ and ∇v are essentially bounded" and "v is Lipschitz" are equivalent.

Remark 1.11 (nodal interpolation). Embeddings into C^0 are what make nodal interpolation (Lecture 2) well defined, since they let us evaluate functions at points. In particular for $d = 2$: $H^2 \hookrightarrow C^0$ since $\operatorname{sob}(H^2) = 2-1 = 1 > 0$, and the borderline space W_1^2 also embeds since $\operatorname{sob}(W_1^2) = 2-2 = 0$.

1.8 Density

Theorem 1.12 (density on Lipschitz domains)**.** *If $\Omega \subset \mathbb{R}^d$ is Lipschitz and $1 \le p < \infty$, then $C^\infty(\overline{\Omega})$ is dense in $W_p^m(\Omega)$: for every $u \in W_p^m(\Omega)$ there exists $u_n \in C^\infty(\overline{\Omega})$ with $\|u - u_n\|_{W_p^m(\Omega)} \to 0$.*

Theorem 1.13 (density on open domains)**.** *If $\Omega \subset \mathbb{R}^d$ is open and $1 \le p < \infty$, then $C^\infty(\Omega) \cap W_p^m(\Omega)$ is dense in $W_p^m(\Omega)$.*

Remark 1.14. Both theorems fail for $p = \infty$: the uniform limit of continuous functions is continuous, but $C^0(\overline{\Omega}) \subsetneq L^\infty(\Omega)$.

1.9 Poincaré–Friedrichs inequalities

Two inequalities of Poincaré type recur throughout these notes. One controls a function by its gradient when the function has zero boundary trace; it is what makes the energy form an inner product on $H_0^1(\Omega)$ and underlies the well-posedness argument of Section 1. The other applies when the function has zero mean; it is the analytic engine of the interpolation theory of Lecture 2. Both express the same idea: modulo the kernel of the gradient (the constants), the full W_p^1 norm is controlled by the gradient alone.

Lemma 1.15 (Poincaré–Friedrichs, zero trace)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded and $1 \le p < \infty$. There is a constant C_Ω, depending only on $\operatorname{diam}\Omega$ and p, such that*

$$\|v\|_{L^p(\Omega)} \le C_\Omega \, \|\nabla v\|_{L^p(\Omega)} \qquad \text{for all } v \in W_{p,0}^1(\Omega),$$

where $W_{p,0}^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $W_p^1(\Omega)$ (so $W_{2,0}^1 = H_0^1$). Consequently $|v|_{W_p^1} = \|\nabla v\|_{L^p}$ is a norm on $W_{p,0}^1(\Omega)$, equivalent to the full W_p^1 norm.

Proof. By density it suffices to prove the bound for $v \in C_c^\infty(\Omega)$; the general case follows by passing to the limit. Since Ω is bounded we may translate so that $\Omega \subset \{x : 0 < x_1 < L\}$ with $L = \operatorname{diam}\Omega$. Extending v by zero outside Ω and using the fundamental theorem of calculus in the x_1 direction,

$$v(x) = \int_0^{x_1} \partial_1 v(t, x')\,\mathrm{d}t, \qquad x = (x_1, x').$$

For $p > 1$, Hölder's inequality gives

$$|v(x)|^p \le x_1^{p-1} \int_0^L |\partial_1 v(t, x')|^p\,\mathrm{d}t \le L^{p-1} \int_0^L |\partial_1 v(t, x')|^p\,\mathrm{d}t$$

(for $p = 1$ the factor x_1^{p-1} is simply absent). Integrating first in $x_1 \in (0, L)$ and then in x',

$$\|v\|_{L^p(\Omega)}^p \le L^p \int_\Omega |\partial_1 v|^p\,\mathrm{d}x \le L^p \, \|\nabla v\|_{L^p(\Omega)}^p,$$

which is the claim with $C_\Omega = \operatorname{diam}\Omega$. The norm equivalence then follows from $\|\nabla v\|_{L^p} \le \|v\|_{W_p^1} \le (1 + C_\Omega)\, \|\nabla v\|_{L^p}$. ∎

The energy inner product. Lemma 1.15 is exactly what makes the Dirichlet form $\mathcal{B}[u,v] = \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x$ an *inner product* on $H_0^1(\Omega)$: bilinearity and symmetry are immediate, and positive definiteness is the implication $\mathcal{B}[v,v] = \|\nabla v\|_{L^2}^2 = 0 \Rightarrow v = 0$, which is precisely the case $C_\Omega \cdot 0 \geq \|v\|_{L^2}$ of the lemma. This is the fact invoked (by forward reference) in the well-posedness argument of Section 1, where it also supplies the bound $\|v\|_{L^2} \leq C_\Omega \|\nabla v\|_{L^2}$ used to show F is bounded.

Lemma 1.16 (generalized Poincaré inequality, zero mean)**.** *Let $B \subset \mathbb{R}^d$ be a bounded, connected, Lipschitz domain and $1 \leq p \leq \infty$. There is a constant $C_P = C_P(B,p)$ such that every $g \in W_p^1(B)$ with vanishing mean satisfies*

$$\int_B g \,\mathrm{d}x = 0 \quad \Longrightarrow \quad \|g\|_{L^p(B)} \leq C_P \, |g|_{W_p^1(B)} = C_P \, \|\nabla g\|_{L^p(B)} .$$

Proof. Argue by contradiction. If no such C_P existed, there would be a sequence $g_n \in W_p^1(B)$ with

$$\int_B g_n \,\mathrm{d}x = 0, \qquad \|g_n\|_{L^p(B)} = 1, \qquad |g_n|_{W_p^1(B)} \to 0.$$

The g_n are then bounded in $W_p^1(B)$, so by the Rellich–Kondrachov theorem (the embedding $W_p^1(B) \hookrightarrow L^p(B)$ is compact for a Lipschitz B) a subsequence, not relabeled, converges in $L^p(B)$ to some g. Since $\nabla g_n \to 0$ in $L^p(B)$ as well, the sequence is Cauchy in $W_p^1(B)$ and the limit obeys $\nabla g = 0$; hence g is constant on the connected set B. Passing to the limit in $\int_B g_n \,\mathrm{d}x = 0$ gives $\int_B g \,\mathrm{d}x = 0$, so that constant is zero, i.e. $g \equiv 0$. This contradicts $\|g\|_{L^p(B)} = \lim \|g_n\|_{L^p(B)} = 1$. □

Remark 1.17 (two faces of one idea). Each lemma removes the kernel of the gradient by a normalization: a boundary condition in Lemma 1.15, a zero-mean condition in Lemma 1.16. The zero-trace form drives coercivity and well-posedness (Lecture 1); the zero-mean form drives polynomial approximation and the interpolation estimates (Lecture 2). The contradiction proof of Lemma 1.16 applies verbatim to Lemma 1.15 (replacing the zero-mean constraint by the zero-trace one); we gave the direct, constant-explicit argument for the latter because it also yields $C_\Omega = \operatorname{diam} \Omega$.

1.10 Extension*

* *This subsection is supplementary and may be skipped on first reading: the extension theorem is not invoked anywhere else in this short course. It is included because extension operators are a fundamental tool in the broader theory of Sobolev spaces—underlying trace theorems, embeddings on general domains, and interpolation of operators.*

Theorem 1.18 (extension)**.** *Let $\Omega \subset \mathbb{R}^d$ be Lipschitz. For every $1 \leq p \leq \infty$ and $m \geq 1$ there exists a bounded linear* extension operator

$$E \colon W_p^m(\Omega) \longrightarrow W_p^m(\mathbb{R}^d)$$

such that $Eu|_\Omega = u$ and $\|Eu\|_{W_p^m(\mathbb{R}^d)} \leq C(\Omega) \, \|u\|_{W_p^m(\Omega)}$.

The proof (Calderón; see Evans for the $k = 1$, $\partial\Omega \in C^1$ case) proceeds by *flattening and reflection*: near each boundary point flatten $\partial\Omega$ to a hyperplane, reflect u across it with coefficients chosen to match the first $m-1$ derivatives, then patch the local extensions with a partition of unity.

Remark 1.19 (necessity of Lipschitz). Extension fails on cusp domains. If $E \colon W_p^1(\Omega) \to W_p^1(\mathbb{R}^d)$ existed for the cusp above, the Sobolev embedding on the surrounding ball would give $W_p^1(\Omega) \hookrightarrow L^\infty(\Omega)$ for $p > 2$, contradicting Example 4.

1.11 Nonsymmetric elliptic problems and the Lax–Milgram theorem

The Riesz representation theorem exploits symmetry: $\mathcal{B}[\cdot,\cdot]$ is itself an inner product on $H_0^1(\Omega)$, and the solution u is simply the Riesz representative of $F = \langle f, \cdot\rangle$. For a *nonsymmetric* problem — e.g. one with a convection term $\boldsymbol{b} \cdot \nabla u$ or a nondivergence-form coefficient matrix $A \neq A^\top$ — the bilinear form is no longer symmetric and this argument breaks down. The correct replacement is the following.

Theorem 1.20 (Lax–Milgram)**.** *Let V be a Hilbert space with norm $\|\cdot\|_V$, let $\mathcal{B}[\cdot,\cdot] : V \times V \to \mathbb{R}$ be a bilinear form satisfying*

$$\begin{aligned} &\textit{(continuity)} \quad |\mathcal{B}[u,v]| \le M\ \|u\|_V \|v\|_V\,, \\ &\textit{(coercivity)} \quad \mathcal{B}[v,v] \ge \alpha\ \|v\|_V^2\,, \end{aligned}$$

for constants $M, \alpha > 0$ independent of $u, v \in V$. Let $F : V \to \mathbb{R}$ be a bounded linear functional with $|F(v)| \le \Lambda\ \|v\|_V$. Then there exists a unique *$u \in V$ such that*

$$\mathcal{B}[u,v] = F(v) \qquad \forall\, v \in V,$$

and the solution satisfies the stability bound $\|u\|_V \le \Lambda/\alpha$.

Proof sketch. Fix $u \in V$. Since $v \mapsto \mathcal{B}[u,v]$ is a bounded linear functional, the Riesz theorem gives a unique $Au \in V$ with $\mathcal{B}[u,v] = \langle Au, v\rangle_V$ for all v; the map $A : V \to V$ is linear and bounded ($\|Au\|_V \le M\,\|u\|_V$). The equation $\mathcal{B}[u,v] = F(v)$ becomes $Au = u_F$ (the Riesz representative of F). Coercivity gives $\alpha\,\|u\|_V^2 \le \mathcal{B}[u,u] = \langle Au, u\rangle_V \le \|Au\|_V\,\|u\|_V$, so $\|Au\|_V \ge \alpha\,\|u\|_V$: A is bounded below and hence injective with closed range. One checks surjectivity (e.g. by a Banach fixed-point iteration $u \leftarrow u - \rho(Au - u_F)$ for $\rho > 0$ small enough), yielding the result. ☐

Remark 1.21. When $\mathcal{B}[\cdot,\cdot]$ is symmetric it defines an inner product equivalent to $\langle\cdot,\cdot\rangle_V$, and A is the identity (up to scaling), so Lax–Milgram reduces to the Riesz theorem. Thus Riesz is the symmetric special case of Lax–Milgram, and for the model Poisson problem of this lecture the simpler Riesz theorem already suffices.

A general elliptic model problem. Consider the *nonsymmetric second-order elliptic problem*

$$\begin{cases} -\operatorname{div}(A\nabla u) + \boldsymbol{b}\cdot\nabla u + cu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{8}$$

where $A \in L^\infty(\Omega;\mathbb{R}^{d\times d})$ is symmetric and uniformly elliptic, $\boldsymbol{b} \in L^\infty(\Omega;\mathbb{R}^d)$, $c \in L^\infty(\Omega)$, and $f \in L^2(\Omega)$. Set $V = H_0^1(\Omega)$ with norm $\|v\|_V = |v|_{H^1} = \|\nabla v\|_{L^2}$ (equivalent by Poincaré). The bilinear form is

$$\mathcal{B}[u,v] = \int_\Omega A\nabla u \cdot \nabla v \,\mathrm{d}x + \int_\Omega (\boldsymbol{b}\cdot\nabla u)\, v \,\mathrm{d}x + \int_\Omega cu\, v\,\mathrm{d}x.$$

Continuity is immediate from Hölder and the L^∞ bounds. *Coercivity* uses the ellipticity condition $A(x)\xi\cdot\xi \ge \lambda\,|\xi|^2$ a.e. and the antisymmetric structure of the convection term: integration by parts gives

$$\int_\Omega (\boldsymbol{b}\cdot\nabla v)\, v\,\mathrm{d}x = -\tfrac{1}{2}\int_\Omega (\operatorname{div}\boldsymbol{b})\, v^2\,\mathrm{d}x,$$

so $\mathcal{B}[v,v] \ge \lambda\,\|\nabla v\|_{L^2}^2 + \int_\Omega (c - \frac{1}{2}\operatorname{div}\boldsymbol{b})\, v^2\,\mathrm{d}x \ge \lambda\,\|\nabla v\|_{L^2}^2$ provided $c - \frac{1}{2}\operatorname{div}\boldsymbol{b} \ge 0$ a.e. Hence the Lax–Milgram theorem applies, giving a unique $u \in H_0^1(\Omega)$ solving (8) weakly.

Remark 1.22. When $\boldsymbol{b} = 0$ and $c = 0$ the bilinear form reduces to $\int_\Omega A\nabla u \cdot \nabla v$, which is symmetric (since $A = A^\top$), and Riesz applies. As soon as $\boldsymbol{b} \neq 0$ the convection term $\int_\Omega (\boldsymbol{b} \cdot \nabla u)\, v$ is not symmetric in u and v, so $\mathcal{B}[u, v] \neq \mathcal{B}[v, u]$ in general and Riesz is no longer available — Lax–Milgram is the right tool. Finite element discretization of (8) proceeds by the Galerkin method exactly as in Lecture 2; Céa's lemma holds with the ratio M/α (not equal to 1 as in the symmetric case), and the approximation theory is unchanged.

2 Discretization

2.1 Galerkin formulation

The Galerkin method replaces the infinite-dimensional space V in the weak problem (3) by a finite-dimensional subspace $V_\mathcal{T} \subset V$: find $U \in V_\mathcal{T}$ such that

$$\mathcal{B}[U, V] = \langle f, V\rangle \qquad \forall\, V \in V_\mathcal{T}. \tag{9}$$

Subtracting (9) from (3) gives *Galerkin orthogonality*

$$\mathcal{B}[u - U, V] = 0 \qquad \forall\, V \in V_\mathcal{T}. \tag{10}$$

For the symmetric, coercive form $\mathcal{B}[\cdot,\cdot]$ this says U is the $\mathcal{B}[\cdot,\cdot]$-orthogonal projection of u onto $V_\mathcal{T}$, i.e. the best approximation in the energy norm $|\!|\!|v|\!|\!| = \mathcal{B}[v, v]^{1/2}$. In general, *Céa's lemma* quantifies this:

$$|u - U|_{H^1_0(\Omega)} \leq \frac{M}{\alpha} \inf_{V \in V_\mathcal{T}} |u - V|_{H^1_0(\Omega)}\,, \tag{11}$$

where M and α are the continuity and coercivity constants. Thus the discretization error is controlled by the *approximation error*, which we estimate next via interpolation.

2.2 Triangulations

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain.

Definition 2.1 (conforming triangulation)**.** A *conforming triangulation* $\mathcal{T}$ of Ω is a finite collection of closed triangles such that

1. $\bar{\Omega} = \bigcup_{T \in \mathcal{T}} T$;
2. the interior of any two distinct triangles is disjoint;
3. any two triangles $T, T' \in \mathcal{T}$ share either the empty set, a common vertex, or a common complete edge (no *hanging nodes*).

The third condition ensures that the triangulation is consistent: edges are not partially shared. Figure 3 shows two configurations it rules out. We write $h_T = \operatorname{diam} T$ for the diameter of T and $h = \max_{T\in\mathcal{T}} h_T$ for the global mesh size.

Definition 2.2 (shape regularity)**.** A triangulation $\mathcal{T}$ is *shape-regular* (or *non-degenerate*) with constant $\sigma > 0$ if

$$\frac{h_T}{\rho_T} \leq \sigma \qquad \forall\, T \in \mathcal{T},$$

where ρ_T is the diameter of the largest ball inscribed in T. A *family* $\{\mathcal{T}\}_h$ of triangulations is shape-regular if σ can be chosen independently of h.

Shape regularity prevents triangles from becoming flat (large angles approaching π) or needle-like (small angles approaching 0). It is the minimal geometric assumption under which interpolation error constants are bounded uniformly in h.

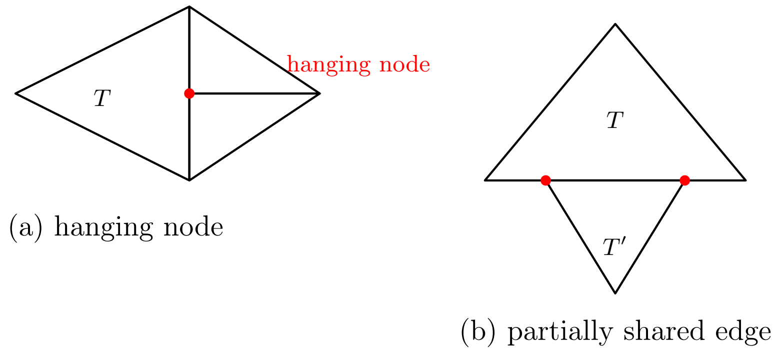


Figure 3: Two situations *excluded* by conformity. (a) A vertex of the two right-hand triangles falls in the interior of an edge of T, creating a *hanging node*. (b) The triangles T, T' share only *part* of an edge, so their intersection is neither a common vertex nor a common complete edge.

2.3 Piecewise linear finite elements ($\mathcal{P}_1$)

Fix a triangle T with vertices z_1, z_2, z_3. The *barycentric coordinates* $\lambda_1, \lambda_2, \lambda_3$ on T are the unique affine functions satisfying $\lambda_i(z_j) = \delta_{ij}$ and $\lambda_1 + \lambda_2 + \lambda_3 = 1$. They provide a partition of unity and are non-negative on T. The space of *piecewise linear polynomials* on T is

$$\mathcal{P}_1(T) = \operatorname{span}\{\lambda_1, \lambda_2, \lambda_3\}, \qquad \dim \mathcal{P}_1(T) = 3.$$

Any $p \in \mathcal{P}_1(T)$ is uniquely determined by its values at the three vertices ($p = \sum_{i=1}^{3} p(z_i)\,\lambda_i$), so the vertex evaluations are a basis of degrees of freedom. The functions $\lambda_1, \lambda_2, \lambda_3$ are themselves the *nodal (hat) basis*: $\lambda_i(z_j) = \delta_{ij}$.

The *global piecewise-linear space* is

$$V_{\mathcal{T}}^1 = \left\{ V \in C^0(\bar{\Omega}) : V|_T \in \mathcal{P}_1(T)\ \forall\, T \in \mathcal{T} \right\}.$$

It has one degree of freedom per mesh node (vertex). The H_0^1-conforming subspace is

$$\mathcal{V}_h = V_{\mathcal{T}}^1 \cap H_0^1(\Omega) = \left\{ V \in V_{\mathcal{T}}^1 : V = 0 \text{ on } \partial\Omega \right\},$$

with basis the hat functions $\{\phi_z\}_{z \in \mathcal{N}_h^{\text{int}}}$ associated to interior nodes $\mathcal{N}_h^{\text{int}}$: ϕ_z is the unique $V \in V_{\mathcal{T}}^1$ with $\phi_z(z) = 1$ and $\phi_z(z') = 0$ at every other node z'. Its support, which we denote

$$\omega_z := \operatorname{supp} \phi_z,$$

is the patch of triangles surrounding z, from now on called the *star* of z (Figure 4); the stars ω_z will be the natural localization patches in the a posteriori theory of Lecture 5. Every $V \in V_{\mathcal{T}}^1$ has the nodal representation $V = \sum_z V(z)\,\phi_z$.

Continuity across edges (and hence H^1-conformity) holds because a $\mathcal{P}_1$ function on a shared edge is determined by its two endpoint values, which are matched by construction.

2.4 Higher-order Lagrange elements ($\mathcal{P}_k$)

For an integer $k \geq 1$, the space of polynomials of total degree at most k on a triangle T is

$$\mathcal{P}_k(T) = \Big\{ p = \sum_{|\alpha| \leq k} c_\alpha x^\alpha : c_\alpha \in \mathbb{R} \Big\}, \qquad \dim \mathcal{P}_k(T) = \binom{k+2}{2} = \frac{(k+1)(k+2)}{2}.$$

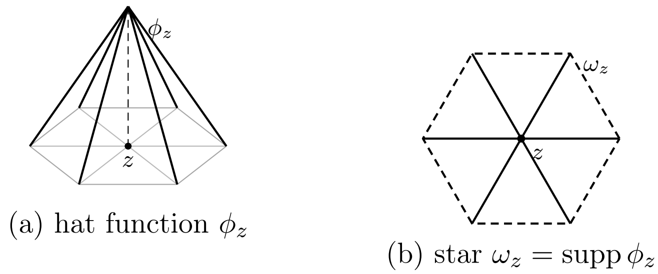


Figure 4: The piecewise-linear nodal basis (hat) function ϕ_z of an interior node z (a), and its support — the star ω_z of triangles containing z (b, boundary dashed, interior skeleton solid). ϕ_z equals 1 at z, 0 at all other nodes, and is linear on each triangle of ω_z.

The natural degrees of freedom are point evaluations at the *principal lattice nodes* of order k,

$$\mathcal{L}_k(T) = \left\{ \tfrac{i}{k} z_1 + \tfrac{j}{k} z_2 + \tfrac{\ell}{k} z_3 : i, j, \ell \geq 0,\ i + j + \ell = k \right\}, \qquad |\mathcal{L}_k(T)| = \dim \mathcal{P}_k(T).$$

These nodes are *unisolvent* on $\mathcal{P}_k(T)$: a polynomial $p \in \mathcal{P}_k(T)$ vanishing at all lattice nodes must be zero. The corresponding nodal basis $\{\phi_\alpha\}$ satisfies $\phi_\alpha(x_\beta) = \delta_{\alpha\beta}$ for $x_\alpha, x_\beta \in \mathcal{L}_k(T)$.

The quadratic case $k = 2$. The six nodes are the three vertices z_1, z_2, z_3 and the three edge midpoints $m_{ij} = \frac{1}{2}(z_i + z_j)$ (Figure 5). In barycentric coordinates the nodal basis functions are

$$\underbrace{N_i = \lambda_i\,(2\lambda_i - 1)}_{\text{vertex } z_i}, \qquad \underbrace{N_{ij} = 4\,\lambda_i \lambda_j}_{\text{midpoint } m_{ij}} \quad (i \neq j).$$

The Kronecker property $\phi_\alpha(x_\beta) = \delta_{\alpha\beta}$ is checked directly: $N_i(z_i) = 1 \cdot (2 - 1) = 1$, while N_i vanishes at z_j ($\lambda_i = 0$) and at every midpoint m_{ij} ($\lambda_i = \frac{1}{2} \Rightarrow 2\lambda_i - 1 = 0$); and $N_{ij}(m_{ij}) = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1$ while N_{ij} vanishes at all other nodes. Note that a vertex function N_i dips negative near the opposite edge, whereas an edge function N_{ij} is an *edge bubble* peaking at m_{ij}.

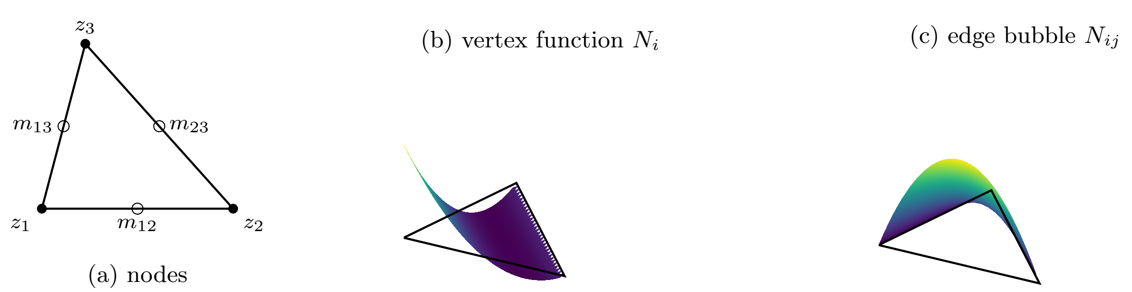


Figure 5: The six $\mathcal{P}_2$ nodes on a triangle (a): vertices z_1, z_2, z_3 (filled) carry the vertex functions $N_i = \lambda_i(2\lambda_i - 1)$, whose graph (b) equals 1 at its own vertex and dips negative near the opposite edge; edge midpoints m_{ij} (open) carry the edge bubbles $N_{ij} = 4\lambda_i\lambda_j$, whose graph (c) peaks at the midpoint and vanishes on ∂T.

These are the $k = 2$ instances of the general $\mathcal{P}_k$ nodal basis: in barycentric coordinates the functions are products of the form $\prod_{i=1}^{3} \prod_{j=0}^{a_i - 1} (\lambda_i - j/k)/((a_i - j)/k)$ for multi-indices (a_1, a_2, a_3) with $a_1 + a_2 + a_3 = k$; for $k = 1$ these reduce to $\lambda_1, \lambda_2, \lambda_3$.

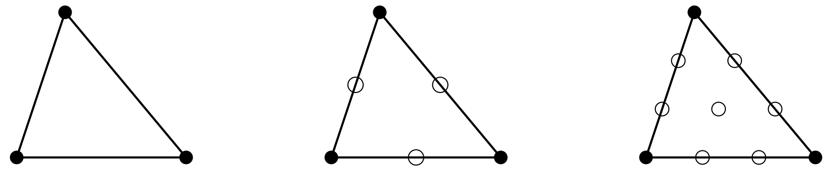


Figure 6: Principal lattice nodes for $\mathcal{P}_1$, $\mathcal{P}_2$, $\mathcal{P}_3$ on a triangle. Filled circles: vertices (shared by all orders). Open circles: edge and interior Lagrange nodes.

The global $\mathcal{P}_k$ space and its H_0^1-conforming subspace are

$$V_{\mathcal{T}}^k = \left\{ V \in C^0(\bar{\Omega}) : V|_T \in \mathcal{P}_k(T)\ \forall T \in \mathcal{T} \right\},$$
$$\mathcal{V}_h = V_{\mathcal{T}}^k \cap H_0^1(\Omega).$$

Global continuity follows because a $\mathcal{P}_k$ polynomial on a shared edge is determined by the $k+1$ edge lattice nodes, which are shared between the two adjacent triangles. For $k=1$ this recovers the $\mathcal{P}_1$ space of the previous subsection.

2.5 Interpolation and a priori error estimates

Define the *nodal (Lagrange) interpolant* $I_{\mathcal{T}} v \in V_{\mathcal{T}}$ by matching v at all nodes; this requires $v \in C^0$, which holds by the embeddings of Lecture 1 when sob is large enough (e.g. $H^2 \hookrightarrow C^0$ in 2D). The cornerstone local estimate compares v with a good *polynomial* approximation on a reference element and then scales back. The polynomial we use is the Verfürth projection, which we introduce first; its key approximation property is a *generalized Poincaré inequality* that will do the analytic work in the proof.

The Verfürth projection. On a bounded, connected Lipschitz domain B (below, the reference element $\hat{T}$), the *Verfürth projection* [1] $P_{k,B} v \in \mathcal{P}_k(B)$ of a function $v \in W_p^{k+1}(B)$ is defined by the moment conditions

$$\int_B D^\alpha \big(v - P_{k,B} v\big)\, \mathrm{d}x = 0 \qquad \text{for all } |\alpha| \le k. \tag{12}$$

Construction and uniqueness. Expanding $q = P_{k,B} v$ about the centroid x_0 of B,

$$q(x) = \sum_{|\alpha| \le k} c_\alpha \frac{(x - x_0)^\alpha}{\alpha!},$$

its $\dim \mathcal{P}_k = \binom{k+d}{d}$ coefficients are fixed by the equally many conditions (12), which are *triangular* in the order $|\alpha|$: since

$$\int_B D^\beta q \, \mathrm{d}x = c_\beta\, |B| \; + \sum_{\alpha > \beta,\ |\alpha| \le k} c_\alpha \int_B \frac{(x - x_0)^{\alpha - \beta}}{(\alpha - \beta)!}\, \mathrm{d}x,$$

the top-order coefficients ($|\beta| = k$, empty sum) are determined first by $c_\beta = \fint_B D^\beta v \, \mathrm{d}x$, and descending in $|\beta|$ fixes the rest uniquely. Taking D^β of this formula also exhibits the commuting-projection property $D^\beta P_{k,B} v = P_{k-|\beta|,B}(D^\beta v)$; in particular $D^{k+1} q = 0$ since $q \in \mathcal{P}_k$.

The reason $P_{k,B}$ is the right comparison polynomial is its approximation property, which is nothing but a higher-order Poincaré inequality.

Lemma 2.3 (Verfürth: a generalized Poincaré inequality)**.** *Let B be a bounded, connected Lipschitz domain, $1 \le p \le \infty$ and $k+1 > d/p$. Then*

$$|v - P_{k,B}v|_{W_p^m(B)} \ \le \ C\, |v|_{W_p^{k+1}(B)} \qquad \textit{for all } 0 \le m \le k,$$

with $C = C(B,k,p)$. For $k = 0$ this is exactly the zero-mean Poincaré inequality of Lemma 1.16; the general case follows by iterating it.

Proof. Write $q = P_{k,B}v$, so $D^{k+1}q = 0$ and, by (12), $\int_B D^\alpha(v-q)\,\mathrm{d}x = 0$ for all $|\alpha| \le k$. Fix a multi-index α with $|\alpha| = k$. Since $D^\alpha(v-q)$ has zero mean, Lemma 1.16 gives

$$\|D^\alpha(v-q)\|_{L^p(B)} \le C\,|D^\alpha(v-q)|_{W_p^1(B)} \le C\,\|D^{k+1}(v-q)\|_{L^p(B)} = C\,|v|_{W_p^{k+1}(B)},$$

the last equality using $D^{k+1}q = 0$. Descending to $|\alpha| = k-1$, the zero-mean condition again applies and $|D^\alpha(v-q)|_{W_p^1(B)} \le \|D^k(v-q)\|_{L^p(B)}$, which was just bounded; iterating down to $|\alpha| = 0$ yields the claim. This iteration of Lemma 1.16 is the sense in which the estimate is a *generalized* Poincaré inequality. ☐

Theorem 2.4 (local interpolation estimate)**.** *For a shape-regular element T and $0 \le m \le k+1$ with $k+1 > d/p$,*

$$|v - I_{\mathcal{T}}v|_{W_p^m(T)} \le C\, h_T^{k+1-m}\ |v|_{W_p^{k+1}(T)}\,.$$

In particular, for $\mathcal{P}_1$ in the energy norm, $\|\nabla(v - I_{\mathcal{T}}v)\|_{L^2(T)} \le C\,h_T\ |v|_{H^2(T)}$.

Proof. The argument has two steps.

Step 1 (Invariance). Work on the reference element $\hat{T}$, with $\hat{\mathcal{I}}$ the Lagrange interpolant there and $q = P_{k,\hat{T}}v$. Because $\hat{\mathcal{I}}$ *reproduces* $\mathcal{P}_k$ exactly ($\hat{\mathcal{I}}q = q$),

$$v - \hat{\mathcal{I}}v = (v-q) - \hat{\mathcal{I}}(v-q).$$

For the interpolated term, boundedness of $\hat{\mathcal{I}} : C^0(\hat{T}) \to \mathcal{P}_k(\hat{T})$, the embedding $W_p^{k+1}(\hat{T}) \hookrightarrow C^0(\hat{T})$, and norm equivalence on the finite-dimensional space $\mathcal{P}_k(\hat{T})$ give

$$|\hat{\mathcal{I}}(v-q)|_{W_p^m(\hat{T})} \lesssim \|v-q\|_{C^0(\hat{T})} \lesssim \|v-q\|_{W_p^{k+1}(\hat{T})}.$$

By Lemma 2.3, both $|v-q|_{W_p^m(\hat{T})}$ and the full norm $\|v-q\|_{W_p^{k+1}(\hat{T})}$ are controlled by $|v|_{W_p^{k+1}(\hat{T})}$ (for the top order use $D^{k+1}q = 0$). Hence

$$|v - \hat{\mathcal{I}}v|_{W_p^m(\hat{T})} \le |v-q|_{W_p^m(\hat{T})} + |\hat{\mathcal{I}}(v-q)|_{W_p^m(\hat{T})} \le C\,|v|_{W_p^{k+1}(\hat{T})}. \tag{13}$$

Step 2 (Scaling). Let $F_T : \hat{T} \to T$ be the affine bijection, $F_T(\hat{x}) = B\hat{x} + b$, with $\|B\| \lesssim h_T$, $\|B^{-1}\| \lesssim h_T^{-1}$, and $|\det B| \sim h_T^d$. For any function w on T set $\hat{w} = w \circ F_T$ on $\hat{T}$. By the chain rule $|D^j\hat{w}(\hat{x})| \lesssim h_T^j |D^j w(F_T(\hat{x}))|$, so a change of variables ($\mathrm{d}x = |\det B|\,\mathrm{d}\hat{x} \sim h_T^d\,\mathrm{d}\hat{x}$) gives

$$\|D^j\hat{w}\|_{L^p(\hat{T})} \lesssim h_T^j\, h_T^{-d/p}\, \|D^j w\|_{L^p(T)},$$

i.e. $\|D^j w\|_{L^p(T)} \lesssim h_T^{d/p-j}\|D^j\hat{w}\|_{L^p(\hat{T})}$. This is exactly the statement that $\mathrm{sob}(W_p^j) = j - d/p$ governs the scaling. Applying this with $w = v - \mathcal{I}_T v$ (and noting $\widehat{v - \mathcal{I}_T v} = \hat{v} - \hat{\mathcal{I}}\hat{v}$) at orders m and $k+1$:

$$\begin{aligned} |v - \mathcal{I}_T v|_{W_p^m(T)} &\lesssim h_T^{d/p-m}\,|\hat{v} - \hat{\mathcal{I}}\hat{v}|_{W_p^m(\hat{T})} \le C\,h_T^{d/p-m}\,|\hat{v}|_{W_p^{k+1}(\hat{T})} && \text{(by (13))} \\ &\lesssim C\,h_T^{d/p-m} \cdot h_T^{-(d/p-(k+1))}\,|v|_{W_p^{k+1}(T)}. \end{aligned}$$

Collecting exponents: $(d/p - m) - (d/p - (k+1)) = k+1-m$, which gives the result. ☐

Summing over $\mathcal{T}$ and combining with Céa's lemma (11) yields the *a priori error estimate.*

Theorem 2.5 (a priori energy estimate)**.** *If $u \in H^{k+1}(\Omega)$ and $V_{\mathcal{T}}$ uses $\mathcal{P}_k$ Lagrange elements on a shape-regular mesh,*

$$|u - U|_{H_0^1(\Omega)} \le C\, h^k \; |u|_{H^{k+1}(\Omega)} \,.$$

A duality (Aubin–Nitsche) argument gives one extra order in L^2: $\|u - U\|_{L^2(\Omega)} \le C\, h^{k+1}\, |u|_{H^{k+1}(\Omega)}$ when the problem is H^2-regular.

Remark 2.6 (curse of low regularity). Estimate 2.5 requires $u \in H^{k+1}$. On a polygon with a reentrant corner the solution behaves like $u \sim r^\gamma$ with $\frac{1}{2} < \gamma < 1$, so $u \in H^{1+s}$ only for $s < \gamma$ and uniform refinement yields the suboptimal rate $h^s \approx N^{-s/2}$. Recovering the optimal rate $N^{-k/2}$ is exactly the motivation for *adaptivity* (Lecture 4 onward).

3 Computational Implementation on a Fixed Mesh

We now turn the Galerkin method of Lecture 2 into a working program for $\mathcal{P}_1$ elements, following a minimal Matlab/Octave code. The mathematics — weak formulation (Lecture 1), hat functions and the Galerkin problem (Lecture 2) — is not repeated; the emphasis is on *how* each object is actually computed. Five tasks organize the code: representing the mesh, assembling the element matrices and load vector, imposing boundary conditions, solving the linear system, and verifying the result against the a priori theory of Lecture 2.

3.1 The model problem

The code solves a mild generalization of the Poisson problem of Lecture 1,

$$-\operatorname{div}(a\nabla u) + c\, u = f \quad \text{in } \Omega, \qquad u = g_D \quad \text{on } \Gamma_D, \qquad a\, \partial_n u = g_N \quad \text{on } \Gamma_N, \tag{14}$$

with $\partial\Omega = \Gamma_D \cup \Gamma_N$, diffusion $a > 0$ and reaction $c \ge 0$ (piecewise constant on the mesh, for simplicity). Testing with functions that vanish on Γ_D and integrating by parts, the Neumann datum enters the right-hand side, and the weak form reads: find $u \in H^1(\Omega)$ with $u = g_D$ on Γ_D such that

$$\int_\Omega a\, \nabla u \cdot \nabla v + \int_\Omega c\, u\, v = \int_\Omega f\, v + \int_{\Gamma_N} g_N\, v \qquad \forall\, v \in H^1_{0,\Gamma_D}(\Omega), \tag{15}$$

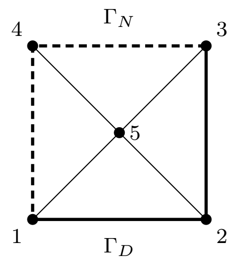
$H^1_{0,\Gamma_D} = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}$. Well-posedness is the Lax–Milgram theory of Lecture 1; the $\mathcal{P}_1$ Galerkin problem replaces H^1 by the space $V^1_{\mathcal{T}}$ of Lecture 2.

3.2 Mesh representation

The triangulation and the boundary partition are stored in four plain-text files:

- `vertex_coordinates.txt` — one line per vertex, its (x, y) coordinates;
- `elem_vertices.txt` — one line per triangle, the three global vertex indices (counterclockwise);
- `dirichlet.txt` — the indices of the vertices lying on Γ_D;
- `neumann.txt` — one line per boundary segment on Γ_N, its two vertex indices.

Figure 7 shows the convention on a four-triangle mesh of the unit square. In the program the first two files are read into arrays `coord` ($N_p \times 2$) and `elem` ($N_t \times 3$); the global $\mathcal{P}_1$ basis $\{\phi_i\}_{i=1}^{N_p}$ is indexed by vertices, one hat function per vertex.




```
vertex_coordinates.txt
 0.0   0.0
 1.0   0.0
 1.0   1.0
 0.0   1.0
 0.5   0.5
```

```
elem_vertices.txt
1   2   5
2   3   5
3   4   5
4   1   5
```

```
dirichlet.txt
 1
 2
 3
```

```
neumann.txt
3   4
4   1
```

Figure 7: Mesh representation on a four-triangle mesh of the unit square (vertex 5 at the centre). Solid edges lie on Γ_D (Dirichlet vertices $1, 2, 3$), dashed edges on Γ_N (segments 3–4 and 4–1). The four text files store the geometry and the boundary partition.

3.3 Reference element and the affine map

Every element computation is pulled back to the reference triangle $\widehat{K} = \{\hat{x} : \hat{x}_1, \hat{x}_2 \geq 0,\ \hat{x}_1 + \hat{x}_2 \leq 1\}$, with vertices $(0, 0), (1, 0), (0, 1)$ and reference hat functions

$$\hat{\phi}_1 = 1 - \hat{x}_1 - \hat{x}_2, \quad \hat{\phi}_2 = \hat{x}_1, \quad \hat{\phi}_3 = \hat{x}_2, \qquad \hat{\nabla}\hat{\phi}_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \ \hat{\nabla}\hat{\phi}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \hat{\nabla}\hat{\phi}_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A triangle T with vertices v^1, v^2, v^3 is the image of $\widehat{K}$ under the affine map (Figure 8)

$$F_T(\hat{x}) = v^1 + B\,\hat{x}, \qquad B = [\, v^2 - v^1 \quad v^3 - v^1 \,] = DF_T, \qquad |T| = \tfrac{1}{2}\,|\det B|\,.$$

The local hat function $\hat{\phi}_i \circ F_T^{-1}$ coincides with the global ϕ_i on T, and by the chain rule the physical and reference gradients are related by

$$\hat{\nabla}\hat{\phi}_i = B^\top \nabla\phi_i, \qquad \text{equivalently} \qquad \nabla\phi_i = B^{-\top}\,\hat{\nabla}\hat{\phi}_i. \tag{16}$$

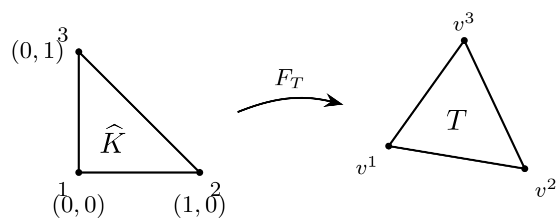


Figure 8: The affine map $F_T(\hat{x}) = v^1 + B\hat{x}$ sends the reference triangle $\widehat{K}$ onto the element T; its Jacobian $B = [\, v^2 - v^1 \ \ v^3 - v^1 \,]$ carries all the geometry, with $|T| = \frac{1}{2}\,|\det B|$ and $\nabla\phi_i = B^{-\top}\hat{\nabla}\hat{\phi}_i$.

3.4 Element matrices and load vector

Because the forms in (15) decouple over elements, it suffices to build 3×3 local matrices. Collect the reference gradients as the columns of $\hat{G} = [\, \hat{\nabla}\hat{\phi}_1 \ \ \hat{\nabla}\hat{\phi}_2 \ \ \hat{\nabla}\hat{\phi}_3 \,] \in \mathbb{R}^{2\times 3}$.

Stiffness. Since B — hence each $\nabla\phi_i$ — is constant on T, (16) gives

$$A^T_{ij} = \int_T a\, \nabla\phi_i \cdot \nabla\phi_j = a\ |T|\ (\hat{\nabla}\hat{\phi}_i)^\top B^{-1} B^{-\top} (\hat{\nabla}\hat{\phi}_j), \qquad \text{i.e.} \qquad A^T = a\ |T|\ \hat{G}^\top \big(B^{-1} B^{-\top}\big) \hat{G}.$$

Mass (reaction). With the exact reference mass matrix,

$$M^T_{ij} = \int_T c\,\phi_i\phi_j = c\,|T|\,\widehat{M}, \qquad \widehat{M} = \frac{1}{12}\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Load. The right-hand side is evaluated with the three-point *midpoint rule*, exact for quadratics,

$$\int_T g\,\mathrm{d}x \;\approx\; \frac{|T|}{3}\big(g(m_{12}) + g(m_{23}) + g(m_{31})\big), \qquad m_{ab} = \tfrac{1}{2}(v^a + v^b),$$

where m_{ab} are the edge midpoints. As each ϕ_i equals $\frac{1}{2}$ at the midpoints of its two adjacent edges and 0 at the opposite one,

$$b^T = \frac{|T|}{3}\begin{pmatrix} \frac{1}{2}\,f(m_{12}) + \frac{1}{2}\,f(m_{31}) \\ \frac{1}{2}\,f(m_{12}) + \frac{1}{2}\,f(m_{23}) \\ \frac{1}{2}\,f(m_{23}) + \frac{1}{2}\,f(m_{31}) \end{pmatrix}.$$

Exercise 3.1 (convection: the non-symmetric case)**.** Extend the code to the full convection–diffusion–reaction problem

$$-\operatorname{div}(a\nabla u) + \boldsymbol{b}\cdot\nabla u + c\,u = f,$$

with a given (piecewise constant on the mesh) velocity field $\boldsymbol{b}$. Show that it adds to every element the *convection matrix*

$$C^T_{ij} = \int_T \phi_i\,(\boldsymbol{b}\cdot\nabla\phi_j)\,\mathrm{d}x = \frac{|T|}{3}\,\boldsymbol{b}\cdot\nabla\phi_j \qquad (i,j = 1,2,3),$$

using that $\nabla\phi_j = B^{-\top}\hat{\nabla}\hat{\phi}_j$ is constant on T and $\int_T \phi_i\,\mathrm{d}x = |T|/3$; in particular the three *rows* of C^T coincide. Add `Cloc = (area/3)*ones(3,1)*(bvec(:)'*Grad)` — with `Grad = B'\Ghat` the physical gradients as columns — to `Aloc` in the assembly loop. Because C^T is *not* symmetric, the global matrix loses symmetry: the remark on Cholesky/conjugate gradients below no longer applies, and one uses a general sparse solver (the backslash still works; iteratively, GMRES or BiCGStab). Verify that for a convection-dominated problem ($|\boldsymbol{b}|\;h \gg a$) the Galerkin solution develops spurious oscillations, whose cure (e.g. SUPG stabilization) lies beyond these notes.

3.5 Global assembly

The global sparse matrix $\mathbf{A}$ ($N_p \times N_p$) and load vector $\mathbf{b}$ are formed by adding each element's contribution into the rows and columns indexed by its global vertices (*scatter-add*). This is the heart of `fem.m`:

```
Ghat = [-1 -1; 1 0; 0 1]'; % reference gradients (columns)
A = sparse(Np, Np); b = zeros(Np, 1);
for T = 1:Nt
    v = elem(T, :); % global indices of the 3 vertices
    P = coord(v, :); % 3x2 vertex coordinates
    B = [P(2,:)-P(1,:); P(3,:)-P(1,:)]'; % Jacobian [v2-v1 v3-v1]
    area = 0.5*abs(det(B));
    m12=(P(1,:)+P(2,:))/2; m23=(P(2,:)+P(3,:))/2; m31=(P(3,:)+P(1,:))/2;
    f12=f(m12); f23=f(m23); f31=f(m31);
    Aloc = a*area*(Ghat'*(B\(B'\Ghat))) ... % a*|T|*Ghat'*inv(B*B')*Ghat
         + c*area*[2 1 1; 1 2 1; 1 1 2]/12; % reaction mass matrix
```

```
    bloc = (area/3)*[ (f12+f31)/2; (f12+f23)/2; (f23+f31)/2 ];
    A(v,v) = A(v,v) + Aloc; % scatter-add
    b(v) = b(v) + bloc;
end
```

The matrix $\mathbf{A}$ is sparse, symmetric, positive (semi-)definite, with $O(N_p)$ nonzeros because each vertex couples only to its mesh neighbours.

3.6 Boundary conditions

Neumann. Each segment $s \subset \Gamma_N$ with endpoints z_a, z_b contributes $\int_s g_N \phi_i$ to its two endpoints. Using Simpson's rule (with $m = \frac{1}{2}(z_a + z_b)$), $\int_s g \approx \frac{|s|}{6}\big(g(z_a) + 4g(m) + g(z_b)\big)$, and ϕ_a, ϕ_b being linear along s,

```
for k = 1:size(neumann,1)
    z = neumann(k,:); za=coord(z(1),:); zb=coord(z(2),:); len=norm(zb-za);
    gm = gN((za+zb)/2);
    b(z) = b(z) + (len/6)*[ gN(za)+2*gm ; 2*gm+gN(zb) ];
end
```

Dirichlet. The values $U_i = g_D(z_i)$ at Dirichlet vertices are imposed by *row replacement*: overwrite row i of $\mathbf{A}$ by $e_i^\top$ and set $b_i = g_D(z_i)$.

```
for i = dirichlet(:)'
    A(i,:) = 0; A(i,i) = 1; b(i) = gD(coord(i,:));
end
U = A \ b; % sparse solve
```

Remark 3.2 (symmetry). Row replacement is the simplest imposition but breaks the symmetry of $\mathbf{A}$. A symmetric alternative eliminates the Dirichlet unknowns (restrict to the free nodes and move the known columns to the right-hand side, $\mathbf{b}_{\text{free}} \mathrel{-}= \mathbf{A}_{\text{free},D}\,\mathbf{g}$), letting one use a sparse Cholesky factorization or conjugate gradients.

3.7 Verification: error and convergence

Correctness is checked with a *manufactured* solution: choose u, set $f = -\operatorname{div}(a\nabla u) + cu$ and $g_D = u|_{\Gamma_D}$, solve, and measure the error. The energy (H^1-seminorm) and L^2 errors are computed element by element with the same midpoint rule; on each T the discrete gradient is the constant vector $\nabla U|_T = B^{-\top}\hat{G}\,U|_T$:

```
gradUT = B' \ (Ghat*U(v)); % constant grad(U) on T
d = gradUT - grad_u(mT); % mT = an edge midpoint
H1 = H1 + (d'*d)/3*area; % accumulate; sum the three midpoints
```

By the a priori estimate of Lecture 2 (Theorem 2.5) a smooth u on a quasi-uniform mesh should give $|u - U|_{H^1} = O(h) = O(N_p^{-1/2})$ and, by the Aubin–Nitsche duality, $\|u - U\|_{L^2} = O(h^2) = O(N_p^{-1})$. Table 1 and Figure 9, for $u(x) = e^{-10|x|^2}$ on the unit square with $a = 1$, $c = 0$ and a sequence of uniform meshes, confirm exactly these rates.

The effect of a reentrant corner. Running the *same* program on the L-shaped domain $\Omega = (-1,1)^2 \setminus \big([0,1] \times [-1,0]\big)$ (`gen_mesh_L_shape`), with the classical reentrant-corner solution

$$u(r,\theta) = r^{2/3}\sin\big(\tfrac{2}{3}\theta\big), \qquad \theta \in [0, \tfrac{3\pi}{2}],$$

N	N_p	$\lvert u-U\rvert_{H^1}$	rate	$\lVert u-U\rVert_{L^2}$	rate
4	25	3.93×10^{-1}	—	3.29×10^{-2}	—
8	81	2.19×10^{-1}	0.85	9.88×10^{-3}	1.73
16	289	1.12×10^{-1}	0.96	2.58×10^{-3}	1.94
32	1089	5.64×10^{-2}	0.99	6.51×10^{-4}	1.98
64	4225	2.83×10^{-2}	1.00	1.63×10^{-4}	2.00
128	16641	1.41×10^{-2}	1.00	4.09×10^{-5}	2.00

Table 1: Convergence for the smooth manufactured solution $u = e^{-10|x|^2}$ on uniform meshes of the unit square. The observed orders in $h = 1/N$ (1 for the energy error, 2 for L^2) match Theorem 2.5.

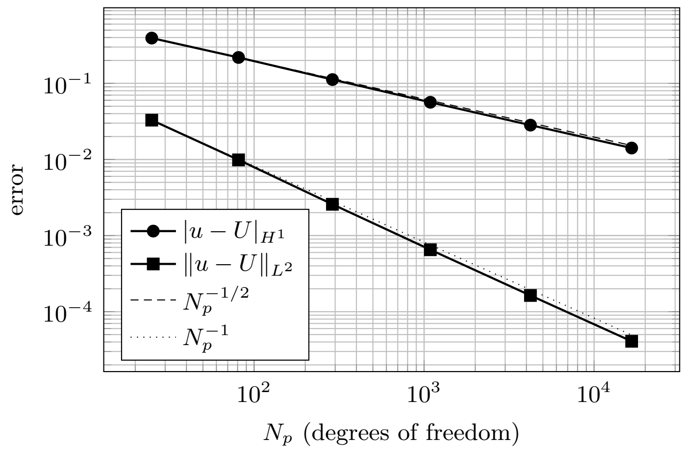


Figure 9: Energy and L^2 errors versus the number of degrees of freedom N_p, for the data of Table 1. The slopes match the reference lines $N_p^{-1/2}$ ($= O(h)$) and N_p^{-1} ($= O(h^2)$).

exposes the limitation of uniform meshes. This u is harmonic (so $f = 0$) and vanishes on the two edges meeting at the corner, but $u \in H^{1+2/3-\varepsilon}$ only, so Theorem 2.5 no longer applies at full order. Table 2 and Figure 10 report the measured errors: the energy error decays like $h^{2/3} = N_p^{-1/3}$ instead of $N_p^{-1/2}$. More strikingly, the L^2 error decays like $h^{4/3} = N_p^{-2/3}$ — *not* the $h^2 = N_p^{-1}$ one might expect. The reason is that the Aubin–Nitsche duality gains only one extra factor $h^{2/3}$ over the energy error, not h: the dual problem inherits the *same* corner singularity and is no more regular than the primal one, so the two singular factors multiply to $h^{2/3} \cdot h^{2/3} = h^{4/3}$.
Recovering the optimal rate $N_p^{-1/2}$ by *local* mesh refinement — without knowing the location or strength of the singularity in advance — is exactly the motivation for the adaptive methods of Lectures 4–9.

3.8 Source code

The driver `fem.m` is shown below in condensed form (mesh input and the Neumann assembly elided); the remaining files are provided as downloads. **Every file is embedded in this PDF**: click a ↓ download link to save it directly (avoiding copy–paste problems). The same files also live in the `fixed-mesh-code` folder distributed with these notes. The Python scripts reproduce the identical $\mathcal{P}_1$ method and were used for Tables 1 and 2.

↓`download the full fem.m`

N	N_p	$\lvert u-U\rvert_{H^1}$	rate	$\lVert u-U\rVert_{L^2}$	rate
4	65	1.82×10^{-1}	—	1.84×10^{-2}	—
8	225	1.17×10^{-1}	0.64	7.24×10^{-3}	1.34
16	833	7.46×10^{-2}	0.65	2.87×10^{-3}	1.34
32	3201	4.74×10^{-2}	0.65	1.14×10^{-3}	1.33
64	12545	3.01×10^{-2}	0.66	4.55×10^{-4}	1.33
128	49665	1.90×10^{-2}	0.66	1.81×10^{-4}	1.33

Table 2: L-shaped domain with the singular solution $u = r^{2/3}\sin(\frac{2}{3}\theta)$ on uniform meshes. The observed orders in $h = 1/N$ are $\frac{2}{3}$ (energy) and $\frac{4}{3}$ (L^2), i.e. $\lvert u-U\rvert_{H^1} = O(N_p^{-1/3})$ and $\lVert u-U\rVert_{L^2} = O(N_p^{-2/3})$ — both suboptimal.

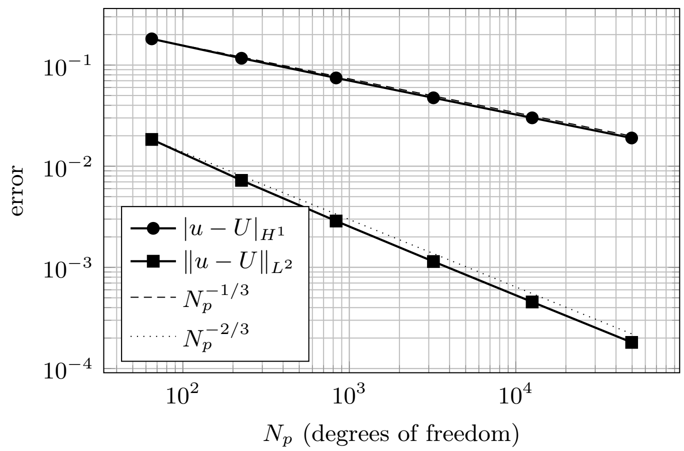


Figure 10: L-shaped domain: energy and L^2 errors versus N_p, for the data of Table 2. The slopes match the reference lines $N_p^{-1/3}$ and $N_p^{-2/3}$; both are below the smooth-case slopes of Figure 9.

```
% fem.m -- solve -div(a grad u) + c u = f with mixed boundary conditions
% (full version, with mesh input and Neumann assembly, available for download)

% ---- problem data ----
coef_a = 1.0; coef_c = 0.0;
fc_f = inline('20*exp(-10*norm(x)^2)*(2-20*norm(x)^2)', 'x');
fc_gD = inline('exp(-10*norm(x)^2)', 'x');

% ---- read mesh: elem_vertices, vertex_coordinates, dirichlet, neumann ----
% ... load the four text files ...
n_vertices = size(vertex_coordinates,1); n_elem = size(elem_vertices,1);
A = sparse(n_vertices, n_vertices); fh = zeros(n_vertices,1);
grd_bas_fcts = [ -1 -1 ; 1 0 ; 0 1 ]'; % reference gradients (columns)

% ---- assembly: loop over elements ----
for el = 1 : n_elem
    v_elem = elem_vertices(el, :);
    v1 = vertex_coordinates(v_elem(1),:)'; v2 = vertex_coordinates(v_elem(2),:)';
    v3 = vertex_coordinates(v_elem(3),:)';
    m12 = (v1+v2)/2; m23 = (v2+v3)/2; m31 = (v3+v1)/2;
```

```
    f12 = fc_f(m12); f23 = fc_f(m23); f31 = fc_f(m31);
    B = [ v2-v1 v3-v1 ]; el_area = abs(det(B))*0.5; Binv = inv(B);
    f_el = [ (f12+f31)*0.5 ; (f12+f23)*0.5 ; (f23+f31)*0.5 ] * (el_area/3);
    el_mat = coef_a * grd_bas_fcts' * (Binv*Binv') * grd_bas_fcts * el_area ...
           + coef_c * el_area * [1/6 1/12 1/12; 1/12 1/6 1/12; 1/12 1/12 1/6];
    fh(v_elem) = fh(v_elem) + f_el;
    A(v_elem, v_elem) = A(v_elem, v_elem) + el_mat;
end

% ... Neumann segments added to fh with Simpson's rule ...

% ---- Dirichlet by row replacement, then solve ----
for i = 1 : length(dirichlet)
    diri = dirichlet(i);
    A(diri,:) = 0; A(diri,diri) = 1; fh(diri) = fc_gD(vertex_coordinates(diri,:));
end
uh = A \ fh;
trimesh(elem_vertices, vertex_coordinates(:,1), vertex_coordinates(:,2), uh);
```

The supporting files (click to save each one):

- ↓gen_mesh_rectangle.m — uniform triangulation of a rectangle (writes the four mesh files).
- ↓gen_mesh_L_shape.m — uniform triangulation of the L-shaped domain.
- ↓H1_err.m — H^1-seminorm error against a known solution.
- ↓L2_err.m — L^2 error against a known solution.
- ↓fem_convergence.py — Python convergence study on the unit square (Table 1).
- ↓fem_convergence_Lshape.py — Python convergence study on the L-shaped domain (Table 2).

4 Adaptive Approximation and Adaptive Mesh Refinement

4.1 Why adapt? Error equidistribution

On a quasi-uniform mesh the convergence rate is limited by the *global* Sobolev regularity of u. Adaptivity instead tailors the mesh to u, and the guiding principle is to *equidistribute* the local error.

A simple one-dimensional situation. The mechanism is already transparent for piecewise-constant approximation on $\Omega = (0,1)$. Let $\mathcal{T} = \{0 = x_0 < x_1 < \cdots < x_N = 1\}$ have subintervals $I_i = (x_{i-1}, x_i)$ of length $h_i = x_i - x_{i-1}$, and approximate a function v by the piecewise constant $v_N(x) = v(x_{i-1})$ for $x \in I_i$ (Figure 11(a)). For $x \in I_i$ the fundamental theorem of calculus gives $v(x) - v_N(x) = \int_{x_{i-1}}^{x} v'(s)\,\mathrm{d}s$, so the accuracy is governed entirely by v'.

Uniform mesh, smooth data. If $v \in W^1_\infty(\Omega)$ and $\mathcal{T}$ is uniform, $h_i = h = 1/N$, then integrating $|v'|$ over I_i gives $|v(x) - v_N(x)| \le h\,\|v'\|_{L^\infty(\Omega)}$, so

$$\|v - v_N\|_{L^\infty(\Omega)} \;\le\; \frac{1}{N}\,|v|_{W^1_\infty(\Omega)}\,. \tag{17}$$

Graded mesh, rough data. Now let v be merely in $W^1_1(\Omega)$, possibly with an unbounded derivative. A uniform mesh no longer delivers (17), but the range of the monotone *error-accumulation function*

$$\phi(x) = \int_0^x \left|v'(s)\right|\,\mathrm{d}s, \qquad \phi(0) = 0, \quad \phi(1) = |v|_{W^1_1(\Omega)}\,,$$

can be cut into N equal pieces.

Example 4.1. For $v(x) = x^\gamma$ with $0 < \gamma < 1$ we have $v'(x) = \gamma x^{\gamma-1} \notin L^\infty(0,1)$, so the uniform bound (17) is unavailable; yet $v \in W_1^1(\Omega)$, and $\phi(x) = \int_0^x \gamma s^{\gamma-1}\,\mathrm{d}s = x^\gamma = v(x)$. The error-accumulation function is the solution itself, and its singular behaviour at the origin is exactly what makes a graded mesh profitable.

Choosing the nodes x_i so that

$$\phi(x_i) = \frac{i}{N}\,|v|_{W_1^1(\Omega)} \qquad (0 \le i \le N)$$

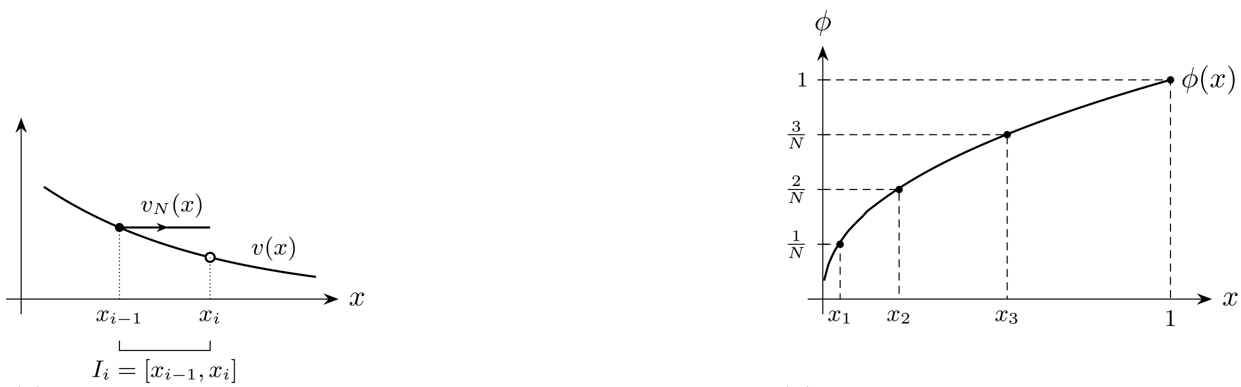
— that is, cutting the *range* of ϕ into N equal pieces (Figure 11(b)) — *equalizes* the local errors: for $x \in I_i$,

$$|v(x) - v_N(x)| \le \int_{x_{i-1}}^{x_i} \left|v'\right|\,\mathrm{d}s = \phi(x_i) - \phi(x_{i-1}) = \frac{1}{N}\,|v|_{W_1^1(\Omega)}\,,$$

whence

$$\|v - v_N\|_{L^\infty(\Omega)} \;\le\; \frac{1}{N}\,|v|_{W_1^1(\Omega)}\,. \tag{18}$$

The graded mesh thus recovers the *same* $O(1/N)$ rate for the rougher class W_1^1 that the uniform mesh gives for W_∞^1. The construction did exactly one thing — it placed the nodes so that every subinterval carries the same portion $1/N$ of the total variation, i.e. it *equidistributed* the local error — and this automatically clusters small elements where $|v'|$ is large (near the singularity of x^γ).



(a) piecewise-constant v_N

(b) equidistributing the range of ϕ

Figure 11: The one-dimensional mechanism. (a) Piecewise-constant approximation v_N on an interval I_i; the error $v - v_N$ is controlled by $\int_{I_i} |v'|$. (b) Grading by equidistribution: cutting the range of the monotone error-accumulation function $\phi(x) = \int_0^x |v'|$ into N equal pieces places the nodes x_i (via $\phi(x_i) = i/N$) so that every subinterval carries the same error $1/N$; the nodes cluster where $|v'|$ is large.

Heuristics for the two-dimensional case. The starting point is the local interpolation estimate written in a form that carries *no explicit power of* h_T. In 2D one has $\mathrm{sob}(W_1^2) = \mathrm{sob}(H^1)$ (both equal 0), yet $W_1^2(\Omega) \hookrightarrow C^0(\bar\Omega)$, so the Lagrange interpolant is well defined and

$$\|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(T)} \;\le\; C\,|u|_{W_1^2(T)} \;=:\; e_T(u,T) \qquad \text{for all } T \in \mathcal{T}. \tag{19}$$

The scale invariance of (19) — the very absence of an h_T factor — is what makes the discrete argument below clean.

A discrete minimization problem. Collect the local errors into the surrogate vector $\mathbf{e} = \big(e_T(u,T)\big)_{T\in\mathcal{T}} \in \mathbb{R}^N$, $N = \#\mathcal{T}$, and consider the total (squared) energy error

$$E_{\mathcal{T}}(u)^2 := \sum_{T\in\mathcal{T}} e_T(u,T)^2.$$

Since the W^2_1-seminorm is an L^1 quantity, it is *exactly additive* over elements, $\sum_{T\in\mathcal{T}} |u|_{W^2_1(T)} = |u|_{W^2_1(\Omega)}$, which ties the local errors to the global regularity through the *constraint*

$$\sum_{T\in\mathcal{T}} e_T(u,T) = C\ |u|_{W^2_1(\Omega)}\,.$$

Among all ways of distributing a fixed total $\sum_T e_T$ over N elements, which minimizes $E_{\mathcal{T}}(u)^2$? Idealizing the e_T to range over all nonnegative reals (shape regularity would couple neighbours, which we ignore), form the Lagrangian

$$\mathcal{L}[\mathbf{e},\lambda] = \sum_{T\in\mathcal{T}} e_T(u,T)^2 - \lambda\Big(\sum_{T\in\mathcal{T}} e_T(u,T) - C\ |u|_{W^2_1(\Omega)}\Big).$$

The optimality condition $\partial\mathcal{L}/\partial e_T = 2\,e_T(u,T) - \lambda = 0$ forces

$$e_T(u,T) = \frac{\lambda}{2} \qquad \text{for every } T\in\mathcal{T},$$

that is, the local error is the *same on every element.* We record the conclusion:

A graded mesh is quasi-optimal if the local error is equidistributed.

The resulting rate. Equidistribution fixes $e_T \equiv \lambda/2$, so the two displays above become

$$E_{\mathcal{T}}(u)^2 = \frac{\lambda^2}{4}\,N, \qquad C\ |u|_{W^2_1(\Omega)} = \frac{\lambda}{2}\,N,$$

and eliminating λ gives the optimal decay rate

$$\boxed{E_{\mathcal{T}}(u) = \|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)} \ \lesssim\ |u|_{W^2_1(\Omega)}\ N^{-1/2}.} \tag{20}$$

This is the same $N^{-1/2}$ that uniform refinement attains for $u \in H^2(\Omega)$, but now requiring only $u \in W^2_1(\Omega)$ — a genuine instance of nonlinear approximation, in which the mesh is designed for the particular u at hand.

Remark 4.2 (discrete vs. continuous equidistribution)*.* The classical route to the same conclusion is *continuous*: introduce a mesh-density $h(x)$, express $\#\mathcal{T} \approx \int_\Omega h^{-2}\,\mathrm{d}x$ and the error via $\int_\Omega h^{2(p-1)}\,\big|D^2u\big|^p\,\mathrm{d}x$, and minimize with a Lagrangian. The resulting Euler–Lagrange equation

$$h(x)^{2p}\ \big|D^2u(x)\big|^p = \text{const} \qquad \text{(Babuška–Rheinboldt for } d=1;\ \text{Nochetto–Veeser for } d=2)$$

expresses the same equidistribution. The discrete formulation above is closer to what an adaptive code actually manipulates — element indicators on a partition — and it needs no mesh-density idealization and no positive power of h_T in (19).

Example 4.3 (W^2_1-regularity at a reentrant corner)**.** Let $u(r,\theta) \sim r^\gamma$ with $0<\gamma<1$, the typical corner singularity. Then $D^2u \sim r^{\gamma-2} \notin L^2$, so $u \notin H^2(\Omega)$; but

$$\|D^2u\|_{L^1(\Omega)} = c\int_0^1 r^{\gamma-2}\cdot r\,dr = c\int_0^1 r^{\gamma-1}\,dr < \infty \implies \boxed{u \in W^2_1(\Omega).}$$

Hence $|u|_{W^2_1(\Omega)} < \infty$, and by (20) a mesh that equidistributes the local error — necessarily *graded* toward the corner — achieves $\|\nabla(u - I_{\mathcal{T}}u)\|_{L^2(\Omega)} \lesssim |u|_{W^2_1(\Omega)}\, N^{-1/2}$, the optimal 2D rate despite $u \notin H^2(\Omega)$.

Remark 4.4 (DeVore diagram)*.* Increasing p from 2 to 1 along the Sobolev line $s - 2/p = \text{const} = \text{sob}(H^1) = 0$ keeps the approximation rate $N^{-1/2}$ fixed while relaxing differentiability requirements; the trade-off is captured by the $(1/p, s)$ diagram below. Nonlinear (adaptive) approximation reaches W^2_1 at the same rate that linear (uniform) approximation reaches H^2.

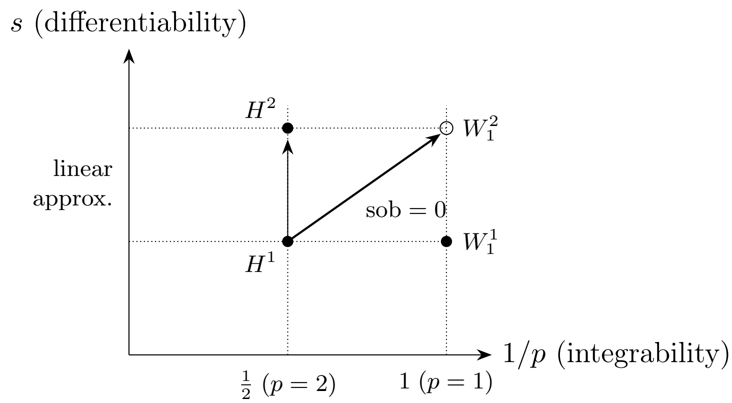


Figure 12: DeVore diagram. The Sobolev line $\text{sob}(W^s_p) = s - 2/p = 0$ (i.e. $\text{sob}(H^1)$) governs nonlinear approximation: graded meshes reach W^2_1 (open circle) at the same $N^{-1/2}$ rate that uniform meshes reach H^2 (filled circle).

The practical question is how to *construct* optimal graded meshes automatically, by local refinement, while preserving conformity and shape regularity. The answer is bisection.

4.2 Bisection

Consider $d = 2$ and *newest-vertex bisection.* Each triangle T carries an ordered triple of edge labels $(i, i+1, i+1)$, where i is the *generation* (level) of T and the level-i edge is the designated *refinement edge.* Bisecting T joins the newest vertex to the midpoint of the refinement edge, producing two children of generation $i+1$ with labels $(i+1, i+2, i+2)$:

$$T \longrightarrow T_1,\, T_2.$$

Properties.

1. Bisection *preserves shape regularity*: all descendants of $\mathcal{T}_0$ fall into finitely many similarity classes, with constants depending only on $\mathcal{T}_0$.

2. The generation of T equals the number of bisections needed to create it.

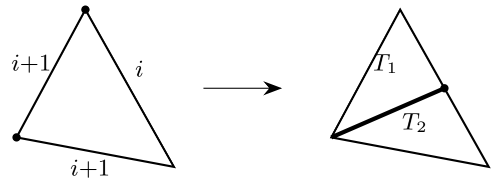


Figure 13: Newest-vertex bisection: T with labels $(i, i{+}1, i{+}1)$ is split across its level-i refinement edge into children T_1, T_2 of generation $i + 1$.

3. A *consistent initial labeling* of $\mathcal{T}_0$ (neighboring triangles agreeing on shared refinement edges) is always achievable after a suitable matching/refinement of $\mathcal{T}_0$, and is what guarantees that refinement terminates and stays conforming.

4.3 Complexity

Bisecting one marked triangle generally creates a hanging node; restoring conformity forces neighboring bisections (*completion*). The propagation can be long, so a naive count of "one split per mark" is wrong. The key result bounds the *total* extra elements by the *total* number of marks, with a constant independent of the refinement history.

Let $\mathcal{M}_j \subset \mathcal{T}_j$ be the marked set at step j and $\mathcal{T}_{j+1} = \textsc{Refine}(\mathcal{T}_j, \mathcal{M}_j)$.

Theorem 4.5 (Binev–Dahmen–DeVore ($d = 2$), Stevenson ($d \geq 3$))**.** *There is a constant C_0, depending only on $\mathcal{T}_0$ and its initial labeling, such that*

$$\#\mathcal{T}_k - \#\mathcal{T}_0 \ \leq\ C_0 \sum_{j=0}^{k-1} \#\mathcal{M}_j.$$

This linear bound is the combinatorial backbone of the optimality theory: it converts a bound on the number of marks (Lecture 9) into a bound on the number of degrees of freedom.

4.4 The greedy algorithm

Given $u \in H^1(\Omega)$, we want to *construct* a near-optimal mesh by iterative refinement. Define the local error $E_T := \|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(T)}$.

Algorithm (Greedy adaptive approximation). Given a tolerance $\delta > 0$ and an initial mesh $\mathcal{T}$:

> **while** $\mathcal{M} := \{T \in \mathcal{T} : E_T > \delta\} \neq \emptyset$**:**
> $\mathcal{T} \leftarrow \textsc{Refine}(\mathcal{T}, \mathcal{M})$
> **return** $\mathcal{T}$

Remark 4.6. This algorithm assumes we know u and can evaluate E_T. It serves as a theoretical benchmark for what adaptive methods can achieve; in practice E_T is replaced by a computable estimator.

Theorem 4.7 (optimal greedy approximation)**.** *If $u \in H^1_0(\Omega) \cap W^2_p(\Omega)$ for some $p > 1$ and $d = 2$, the output $\mathcal{T}$ of $\textsc{Greedy}(\mathcal{T}_0, \delta)$ satisfies*

1. $\|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(\Omega)} \leq \delta\,(\#\mathcal{T})^{1/2}$;

2. $\#\mathcal{T} - \#\mathcal{T}_0 \lesssim \delta^{-1}\ |\Omega|^{1-1/p}\ |u|_{W^2_p(\Omega)}$.

In particular, with $N = \#\mathcal{T} \geq 2\#\mathcal{T}_0$,

$$\|\nabla(u - I_{\mathcal{T}} u)\|_{L^2(\Omega)} \lesssim |\Omega|^{1-1/p} \, |u|_{W_p^2(\Omega)} \, N^{-1/2}.$$

Proof. We proceed in five steps.

1. Local error bound. By the interpolation estimate, $E_T \leq C\, h_T^r \, |u|_{W_p^2(T)}$ with $r = 2(1-1/p) > 0$, so the algorithm terminates. At termination $E_T \leq \delta$ for all $T \in \mathcal{T}$, giving

$$\|\nabla(u - I_{\mathcal{T}} u)\|^2_{L^2(\Omega)} = \sum_{T \in \mathcal{T}} E_T^2 \leq \delta^2 \, \#\mathcal{T},$$

which is statement (1).

2. Counting marked elements. Let $\mathcal{M}$ be the *total* set of all elements ever marked. By the complexity theorem (Theorem 4.5), $\#\mathcal{T} - \#\mathcal{T}_0 \leq C_0 \, \#\mathcal{M}$, so it suffices to bound $\#\mathcal{M}$. Partition $\mathcal{M}$ by element size: let P_j be the marked elements with area $2^{-(j+1)} \leq |T| < 2^{-j}$, $j \geq 0$; then $\#\mathcal{M} = \sum_{j \geq 0} \#P_j$.

3. Bounding $\#P_j$ (geometry). Elements in P_j are disjoint and each has area $\geq 2^{-(j+1)}$, so

$$\#P_j \leq |\Omega| \; 2^{j+1}. \tag{21}$$

4. Bounding $\#P_j$ (regularity). Every $T \in P_j$ was marked because $E_T > \delta$, and $h_T \leq 2^{-j/2}$ (from $|T| < 2^{-j}$), so

$$\delta < E_T \lesssim h_T^r \, |u|_{W_p^2(T)} \lesssim 2^{-jr/2} \, |u|_{W_p^2(T)} \, .$$

Raising to the p-th power and summing over $T \in P_j$,

$$\delta^p \, \#P_j \lesssim 2^{-jrp/2} \, |u|^p_{W_p^2(\Omega)} \, . \tag{22}$$

5. Crossover and summation. Bounds (21) and (22) are complementary; they cross at an index j_0 satisfying $2^{j_0} \approx \delta^{-1} \, |u|_{W_p^2} \, |\Omega|^{-1/p}$. Using the geometric bound for $j < j_0$ and the Sobolev bound for $j \geq j_0$, both geometric series sum to $O(2^{j_0})$, giving

$$\#\mathcal{M} \lesssim |\Omega| \; 2^{j_0} \approx \delta^{-1} \, |u|_{W_p^2(\Omega)} \; |\Omega|^{1-1/p} \, ,$$

which is statement (2). □

5 A posteriori error analysis

Adaptivity needs *computable* error information. We derive an estimator $\mathcal{E}_{\mathcal{T}}(U)$ depending only on the data (f, A, Ω) and the computed solution U, and show it is *two-sided equivalent* to the true energy error. The upper bound (*reliability*) prevents under-refinement; the lower bound (*efficiency*), valid up to a data *oscillation* term, prevents over-refinement and underpins optimality.

Model problem and residual

Consider

$$\begin{cases} -\operatorname{div}(A(x)\nabla u) = f & \text{in } \Omega \subset \mathbb{R}^d, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

with weak solution $u \in V = H_0^1(\Omega)$ and Galerkin solution $U \in V(\mathcal{T})$ (C^0 piecewise linears). The bilinear form $\mathcal{B}[u, v] = \int_\Omega A \nabla u \cdot \nabla v$ is coercive ($\geq \alpha \, |v|^2$) and bounded ($\leq M \, |u| \, |v|$).

Goal. Estimate $|u-U|_{H_0^1(\Omega)}$ purely in terms of computable quantities depending on f, A, Ω, and the Galerkin solution U:

$$\boxed{|u-U|_{H_0^1(\Omega)} \le C\,\mathcal{E}_{\mathcal{T}}(U),}$$

where $\mathcal{E}_{\mathcal{T}}(U)$ is the *a posteriori error estimator.*

Define the *residual* $R \in H^{-1}(\Omega)$ by

$$\langle R, v\rangle = \langle f, v\rangle - \mathcal{B}[U,v] = \mathcal{B}[u-U,v] \qquad \forall\, v \in V = H_0^1(\Omega).$$

5.1 Error–residual relation

The energy error is equivalent to the dual norm of the residual:

$$\alpha\ |u-U|_{H_0^1(\Omega)} \le \|R\|_{H^{-1}(\Omega)} \le M\ |u-U|_{H_0^1(\Omega)}\,.$$

Both bounds follow directly. For the *left inequality*, test the residual against $v = u-U$:

$$\alpha\,|u-U|_{H_0^1}^2 \le \mathcal{B}[u-U,u-U] = \langle R, u-U\rangle \le \|R\|_{H^{-1}}\,|u-U|_{H_0^1}\,,$$

and divide by $|u-U|_{H_0^1}$. For the *right inequality*, use continuity:

$$|\langle R,v\rangle| = |\mathcal{B}[u-U,v]| \le M\,|u-U|_{H_0^1}\,|v|_{H_0^1}\,,$$

so $\|R\|_{H^{-1}} = \sup_v |\langle R,v\rangle| \,/\, |v|_{H_0^1} \le M\,|u-U|_{H_0^1}$.

Thus $|u-U|_{H_0^1} \simeq \|R\|_{H^{-1}}$. It remains to bound $\|R\|_{H^{-1}}$ by computable quantities.

5.2 Residual representation

Assume $A(x) \in C^{0,1}(\bar{\Omega};\mathcal{T}_0)$ (piecewise Lipschitz) and $f \in L^2(\Omega)$. Integrating by parts element by element,

$$\begin{aligned}
\langle R, v\rangle &= \sum_{T\in\mathcal{T}} \int_T fv - \int_T A(x)\nabla U\cdot\nabla v \\
&= \sum_{T\in\mathcal{T}} \int_T \big[f + \operatorname{div}(A\nabla U)\big]v - \int_{\partial T} A\nabla U\cdot\nu_T\, v.
\end{aligned}$$

After cancellation over interior edges, the boundary contributions survive only as *flux jumps.* For two elements T, T' sharing an interior edge s with $\nu_{T'} = -\nu_T$ (see Figure 14), define the *jump residual*

$$j_s := -\big(A_T\nabla U_T\cdot\nu_T + A_{T'}\nabla U_{T'}\cdot\nu_{T'}\big) = [\![\, A\nabla U\cdot\boldsymbol{n}\,]\!]_s,$$

and the *interior (element) residual* $r_T := f + \operatorname{div}(A\nabla U)\big|_T$. Then

$$\langle R, v\rangle = \sum_{T\in\mathcal{T}} \int_T r_T\, v + \sum_{s\in\Gamma}\int_s j_s\, v,$$

where Γ is the mesh skeleton. For $\mathcal{P}_1$ with constant A, $\operatorname{div}(A\nabla U) = 0$ on each T, so $r_T = f|_T$.

Goal. We want to bound $\|R\|_{H^{-1}}$ without inverting any differential operator. The key is *Galerkin orthogonality*: since $\langle R, I_{\mathcal{T}}v\rangle = \mathcal{B}[u-U, I_{\mathcal{T}}v] = 0$, we may replace v by $v - I_{\mathcal{T}}v$ for *free*:

$$\langle R, v\rangle = \langle R, v - I_{\mathcal{T}}v\rangle \qquad \forall\, v\in V.$$

Local interpolation estimates then localize the error to element patches.

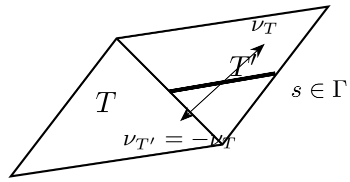


Figure 14: Two elements T, T' sharing interior edge s. Outward normals satisfy $\nu_{T'} = -\nu_T$; the flux jump $j_s = [\![A\nabla U \cdot \nu]\!]$ measures the discontinuity in the normal component of $A\nabla U$ across s.

5.3 Reliability upper bound

Theorem 5.1 (reliability)**.** *There is a constant C depending only on shape regularity such that*

$$|u - U|_{H_0^1(\Omega)} \leq C\, \mathcal{E}_{\mathcal{T}}(U), \qquad \mathcal{E}_{\mathcal{T}}(U) = \Big(\sum_{T \in \mathcal{T}} \mathcal{E}_{\mathcal{T}}^2(U, T) \Big)^{1/2},$$

with the local element indicator $\mathcal{E}_{\mathcal{T}}^2(U,T) = h_T^2 \, \|r_T\|_{L^2(T)}^2 + h_T \, \|j\|_{L^2(\partial T)}^2$.

Proof 1 (quasi-interpolant / Clément–Scott–Zhang). By the error-residual equivalence it suffices to bound $\|R\|_{H^{-1}}$. By Galerkin orthogonality, $\langle R, v \rangle = \langle R, v - I_{\mathcal{T}} v \rangle$ for any $v \in V$. Using the residual representation and Cauchy–Schwarz,

$$\begin{aligned} |\langle R, v \rangle| &= \Big| \sum_T \int_T r_T (v - I_{\mathcal{T}} v) + \sum_s \int_s j_s (v - I_{\mathcal{T}} v) \Big| \\ &\leq \sum_T \|r_T\|_{L^2(T)} \, \|v - I_{\mathcal{T}} v\|_{L^2(T)} + \sum_s \|j_s\|_{L^2(s)} \, \|v - I_{\mathcal{T}} v\|_{L^2(s)} . \end{aligned}$$

The quasi-interpolant satisfies the local estimates

$$\|v - I_{\mathcal{T}} v\|_{L^2(T)} \lesssim h_T \, \|\nabla v\|_{L^2(\omega_T)} , \qquad \|v - I_{\mathcal{T}} v\|_{L^2(s)} \lesssim h_s^{1/2} \, \|\nabla v\|_{L^2(\omega_s)} .$$

Substituting and applying Cauchy–Schwarz again,

$$|\langle R, v \rangle| \lesssim \underbrace{\Big(\sum_T h_T^2 \, \|r_T\|_{L^2(T)}^2 + \sum_s h_s \, \|j_s\|_{L^2(s)}^2 \Big)^{1/2}}_{= \mathcal{E}_{\mathcal{T}}(U)} \underbrace{\Big(\sum_T \|\nabla v\|_{L^2(\omega_T)}^2 \Big)^{1/2}}_{\lesssim \|\nabla v\|_{L^2(\Omega)}},$$

where finite overlap of patches gives the last bound. Dividing by $\|\nabla v\|_{L^2(\Omega)}$ and taking the supremum yields the claim. □

Proof 2 (partition of unity). Let $\{\phi_z\}_{z \in \mathcal{N}}$ be the hat-function partition of unity, $\sum_z \phi_z = 1$. Since $\phi_z \in V(\mathcal{T})$, Galerkin orthogonality gives $\langle R, \phi_z \rangle = 0$ for each z, so

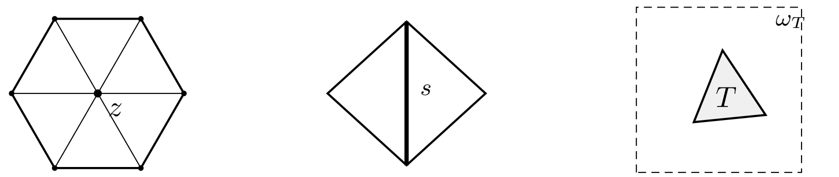
$$\langle R, v \rangle = \sum_z \langle R, v\phi_z \rangle = \sum_z \langle R, (v - \overline{v}_z)\phi_z \rangle , \qquad \overline{v}_z = \fint_{\omega_z} v.$$

Expanding the residual on each star ω_z and applying the Poincaré inequalities

$$\|v - \overline{v}_z\|_{L^2(\omega_z)} \lesssim h_z \, \|\nabla v\|_{L^2(\omega_z)} , \qquad \|v - \overline{v}_z\|_{L^2(\partial\omega_z)} \lesssim h_z^{1/2} \, \|\nabla v\|_{L^2(\omega_z)} ,$$

and Cauchy–Schwarz over stars, we arrive at the same estimate as Proof 1. □

Remark 5.2. Reliability makes the estimator a legitimate stopping criterion: it guarantees that driving $\mathcal{E}_{\mathcal{T}}(U)$ to zero forces the true error to zero. The element indicators $\mathcal{E}_{\mathcal{T}}^2(U,T)$ are exactly what Mark uses in Lecture 6 to decide which elements to refine.



ω_z (node star) ω_s (edge patch) ω_T (neighborhood)

Figure 15: Local patches: node star ω_z (used in Proof 2), edge patch ω_s, and element neighborhood ω_T (used in Proof 1). Finite overlap of patches ensures the $\|\nabla v\|$ factors recombine globally.

5.4 Efficiency: bubble functions and the interior residual

Reliability prevents *under*-refinement. To prevent *over*-refinement — and to make adaptivity optimal — we need the converse: the estimator must not exceed the true error too much (*efficiency*). The price is a data *oscillation* term.

Fix $T \in \mathcal{T}$ and let $r = r_T$ be the interior residual. Let b_T be the cubic *element bubble* ($b_T > 0$ in $\operatorname{int}(T)$, $b_T = 0$ on ∂T, $b_T(x_T) = 1$).

For a *constant* residual r on T, equivalence of norms on the finite-dimensional space $\mathcal{P}_0(T)$ gives $\int_T r^2 \approx \int_T r^2 b_T$. Testing against $v = rb_T \in H_0^1(T)$,

$$\|r\|^2_{L^2(T)} \lesssim \int_T r^2 b_T = \langle r, rb_T\rangle \le \|r\|_{H^{-1}(T)} \|rb_T\|_{H^1(T)} \lesssim \|r\|_{H^{-1}(T)}\, h_T^{-1} \|r\|_{L^2(T)} ,$$

where $|\nabla b_T| \lesssim h_T^{-1}$ was used. Therefore

$$\boxed{h_T \|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} .}$$

Since $\langle r_T, v\rangle = \mathcal{B}[u - U, v]$ for $v \in H_0^1(T)$ (check from IBP), we get $\|r_T\|_{H^{-1}(T)} \le \|\nabla(u - U)\|_{L^2(T)}$, which is the interior lower bound. An analogous *edge bubble* argument on the patch ω_s controls the jump term.

5.5 Oscillation and the full lower bound

The bubble argument is exact only when r is constant. For general data, one splits $r = \overline{r}_T + (r - \overline{r}_T)$ with $\overline{r}_T = ⨍_T r$. The constant part obeys the equivalence above; the remainder contributes the *oscillation terms*

$$\operatorname{osc}_{\mathcal{T}}(U, T)^2 = h_T^2 \|r - \overline{r}_T\|^2_{L^2(T)} + h_T \left\|j - \overline{j}\right\|^2_{L^2(\partial T)} ,$$

which vanish for piecewise-constant data and are of higher order when f is smooth.

Lemma 5.3 (local bound for interior residual)**.**

$$h_T \|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} + h_T \|r - \overline{r}_T\|_{L^2(T)} .$$

Proof. Write $\|h_T r\|_{L^2(T)} \le h_T \|\overline{r}_T\|_{L^2(T)} + h_T \|r - \overline{r}_T\|_{L^2(T)}$. For the first term, apply the constant-residual equivalence to $\overline{r}_T$: $h_T \|\overline{r}_T\|_{L^2(T)} \lesssim \|\overline{r}_T\|_{H^{-1}(T)} \lesssim \|r\|_{H^{-1}(T)} + \|r - \overline{r}_T\|_{H^{-1}(T)}$. It remains to bound $\|r - \overline{r}_T\|_{H^{-1}(T)}$. For any $v \in H_0^1(T)$,

$$|\langle r - \overline{r}_T, v\rangle| = |\langle r - \overline{r}_T, v - \overline{v}_T\rangle| \le \|r - \overline{r}_T\|_{L^2(T)} \|v - \overline{v}_T\|_{L^2(T)} \lesssim h_T \|r - \overline{r}_T\|_{L^2(T)} \|\nabla v\|_{L^2(T)} ,$$

by Poincaré, so $\|r - \overline{r}_T\|_{H^{-1}(T)} \lesssim h_T \|r - \overline{r}_T\|_{L^2(T)}$. Combining gives the lemma. $\square$

Remark 5.4 (interior node property). When $\mathcal{T}_* \geq \mathcal{T}$ adds an interior node $z \in T$, the hat function $\phi_z \in V(\mathcal{T}_*)$ satisfies $\int_T \phi_z \approx |T|$ and can replace the bubble. Testing $\langle r_T, \phi_z \rangle$ and using Galerkin orthogonality with the new solution U_*,

$$\langle r_T, \phi_z \rangle = \mathcal{B}[u - U, \phi_z] = \mathcal{B}[U_* - U, \phi_z]\,,$$

one obtains

$$\boxed{h_T \|r_T\|_{L^2(T)} \lesssim \|\nabla(U_* - U)\|_{L^2(\omega_T)} + \mathrm{osc}_{\mathcal{T}}(U, \omega_T).}$$

This is the *interior node property*: refining T (adding an interior node) directly reduces the estimator, providing the key link between the marked set and the energy decrement used in the contraction proof of Lecture 8.

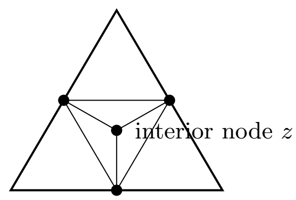


Figure 16: Interior node property: bisecting T (and neighbors for conformity) introduces an interior node $z \in T$. The associated hat function $\phi_z \in V(\mathcal{T}_*)$ acts as a localized test function, showing that the marked elements contribute to the energy decrement.

Theorem 5.5 (efficiency / local lower bound)**.** *For each $T \in \mathcal{T}$, with ω_T the neighborhood of T,*

$$\mathcal{E}_{\mathcal{T}}(U, T) = \Big(h_T^2 \|r\|_{L^2(T)}^2 + h_T \|j\|_{L^2(\partial T)}^2\Big)^{1/2} \;\lesssim\; \|\nabla(u - U)\|_{L^2(\omega_T)} + \mathrm{osc}_{\mathcal{T}}(U, \omega_T).$$

Globally, $C_2\, \mathcal{E}_{\mathcal{T}}^2(U) \leq |u - U|_{H_0^1(\Omega)}^2 + \mathrm{osc}_{\mathcal{T}}^2(U)$.

Sketch. For the interior term: Lemma 5.3 gives $h_T \|r\|_{L^2(T)} \lesssim \|r\|_{H^{-1}(T)} + \mathrm{osc}$, and $\|r\|_{H^{-1}(T)} \leq \|\nabla(u - U)\|_{L^2(T)}$ via the residual identity. For the jump term: an edge bubble b_s supported on ω_s serves as test function and the analogous argument gives $h_s^{1/2} \|j\|_{L^2(s)} \lesssim \|\nabla(u - U)\|_{L^2(\omega_s)} + \mathrm{osc}$. Squaring, summing over T, and using finite overlap gives the global bound. □

Remark 5.6 (two-sided control). Theorems 5.1 and 5.5 together give

$$C_2\, \mathcal{E}_{\mathcal{T}}^2(U) - \mathrm{osc}_{\mathcal{T}}^2(U) \;\leq\; |u - U|_{H_0^1(\Omega)}^2 \;\leq\; C_1\, \mathcal{E}_{\mathcal{T}}^2(U).$$

The estimator is equivalent to the energy error up to oscillation. Since oscillation is reduced by refinement and is generically higher-order, $\mathcal{E}_{\mathcal{T}}(U)$ is a faithful, computable proxy for the energy error — the foundation for the convergence and optimality results of Lectures 8–9.

6 Implementation of an Adaptive Method

We assemble the pieces into the standard adaptive loop and discuss the data structures and algorithms behind each module.

6.1 The adaptive loop

Algorithm (AFEM). Given $\mathcal{T}_0$ and tolerance `tol`, set $k = 0$ and iterate

$U_k = \text{SOLVE}(\mathcal{T}_k)$;
$\{\mathcal{E}_k(U_k, T)\}_{T\in\mathcal{T}_k} = \text{ESTIMATE}(U_k, \mathcal{T}_k)$;
if $\mathcal{E}_k(U_k) \leq$ `tol` **stop**;
$\mathcal{M}_k = \text{MARK}\big(\{\mathcal{E}_k(U_k, T)\}, \theta\big)$;
$\mathcal{T}_{k+1} = \text{REFINE}(\mathcal{T}_k, \mathcal{M}_k)$;
$k \leftarrow k + 1$.

6.2 SOLVE and ESTIMATE

SOLVE is the fixed-mesh solver of Lecture 3 applied to $\mathcal{T}_k$. ESTIMATE computes the element indicators of Theorem 5.1. For $\mathcal{P}_1$ and piecewise constant A, the interior residual is $r_T = f|_T$ and the jump term collects normal-flux jumps across the three edges of T:

```
function eta = estimate(coord, elem, edge, U, f)
  Nt = size(elem,1); eta = zeros(Nt,1);
  gradU = elementGradients(coord, elem, U); % constant grad per element (Lec 3)
  for T = 1:Nt
    P = coord(elem(T,:),:);
    hT = elementDiameter(P);
    aT = elementArea(P);
    % interior residual term hT^2 * ||f||_{L2(T)}^2 (centroid quadrature)
    rT2 = hT^2 * aT * f(mean(P))^2;
    % jump term hT * sum over edges ||[grad U . n]||_{L2(edge)}^2
    jT2 = 0;
    for e = edgesOf(T)
      if isInterior(e)
        jump = (gradU(T,:) - gradU(neighbor(T,e),:)) * normal(e);
        jT2 = jT2 + hT * edgeLength(e) * jump^2;
      end
    end
    eta(T) = sqrt(rT2 + jT2);
  end
end
```

Computing jumps requires an *edge–element adjacency* structure: an `edge` array listing each unique edge once together with its (one or two) incident triangles. This is built in $O(N_t)$ by sorting vertex pairs.

6.3 MARK: Dörfler (bulk) marking

We select a minimal-cardinality set $\mathcal{M}_k$ capturing a fixed fraction θ of the total estimated error:

$$\mathcal{E}_k(U_k, \mathcal{M}_k) \;\geq\; \theta\, \mathcal{E}_k(U_k, \mathcal{T}_k), \qquad 0 < \theta < 1. \tag{23}$$

In practice: sort the indicators in decreasing order and accumulate until the squared partial sum reaches θ^2 of the total.

```
function M = mark(eta, theta)
  [sorted, idx] = sort(eta.^2, 'descend');
  total = sum(sorted);
```

```
  csum = cumsum(sorted);
  k = find(csum >= theta^2 * total, 1, 'first'); % minimal #marked
  M = idx(1:k);
end
```

The exact sort costs $O(N_t \log N_t)$; an approximate binning (quantile) selection achieves the same bulk fraction in $O(N_t)$ and preserves the optimality guarantees of Lecture 9.

6.4 Refine: bisection with completion

Refine bisects every marked triangle along its refinement edge and then runs a *completion* loop that bisects any triangle with a hanging node on its refinement edge, repeating until the mesh is conforming (Lecture 4). Newest-vertex labeling makes the refinement edge a property of the element, so completion is a local, terminating queue process.

```
function [coord, elem] = refine(coord, elem, M)
  markEdge = refinementEdges(elem(M,:)); % tag refinement edges of marked elems
  do
     newHang = neighborsWithHangingNode(elem, markEdge); % conformity violations
     markEdge = markEdge UNION refinementEdges(newHang);
  while newHang nonempty % completion: propagate until conforming
  [coord, elem] = bisectAll(coord, elem, markEdge); % insert midpoints, split
end
```

By Theorem 4.5 the completion adds only $O(\#\mathcal{M}_k)$ elements in total over the whole run, so the loop is not a bottleneck.

6.5 Practical notes

- **Warm starts.** Interpolate U_k onto $\mathcal{T}_{k+1}$ to initialize the iterative solver; nested meshes make this an exact, local operation.
- **Choice of θ.** Small θ refines aggressively (fewer iterations, more DOFs per step); θ near 1 refines almost everything (close to uniform). Values $\theta \in [0.3, 0.6]$ are typical and, by Lecture 9, any $\theta < \theta_* = \sqrt{C_2/C_1}$ yields optimal rates.
- **Validation.** On an L-shaped domain with the singular solution $u \sim r^{2/3}$, the energy error of the adaptive run should recover the optimal slope $N^{-1/2}$, versus the suboptimal $N^{-1/3}$ of uniform refinement — the headline payoff of the whole method.

7 Basic Properties of AFEM

Before proving convergence we assemble the structural properties of the adaptive loop on which the contraction argument (Lecture 8) rests. Two of them — the *localized upper bound* and the *estimator reduction* — quantify how a single Solve–Estimate–Mark–Refine step changes the discrete solution and the estimator.

7.1 The adaptive loop

The loop of Lecture 6 produces a sequence of nested conforming meshes

$$\mathcal{T}_0 \le \mathcal{T}_1 \le \mathcal{T}_2 \le \cdots, \qquad V_k := V(\mathcal{T}_k) \subset V(\mathcal{T}_{k+1}) =: V_{k+1},$$

with Galerkin solutions $U_k \in \mathbb{V}_k$, element indicators $\mathcal{E}_k(U_k, T)$, and marked sets $\mathcal{M}_k \subset \mathcal{T}_k$ selected by Dörfler marking (23). Nestedness is the structural consequence of refinement by bisection: no element is ever coarsened, so $\mathbb{V}_k \subset \mathbb{V}_{k+1}$ and the discrete spaces form an increasing hierarchy. Throughout the analysis $|||v|||_\Omega = \mathcal{B}[v, v]^{1/2}$ denotes the energy norm (equivalent to $|\cdot|_{H_0^1(\Omega)}$), and we abbreviate

$$e_k = |||u - U_k|||_\Omega, \qquad E_k = |||U_{k+1} - U_k|||_\Omega, \qquad \mathcal{E}_k = \mathcal{E}_k(U_k, \mathcal{T}_k), \qquad \mathrm{osc}_k = \mathrm{osc}_k(U_k).$$

The single-step increment $U_{k+1} - U_k \in \mathbb{V}_{k+1}$ plays a central role below.

7.2 Building blocks

We collect the structural properties established in Lecture 5, specialized to the nested meshes $\mathcal{T}_k \leq \mathcal{T}_{k+1}$:

1. **Upper bound** (reliability, Thm. 5.1): $e_k^2 \leq C_1\, \mathcal{E}_k^2$.
2. **Lower bound** (efficiency, Thm. 5.5): $C_2\, \mathcal{E}_k^2 \leq e_k^2 + \mathrm{osc}_k^2$.
3. **Localized upper bound** (Lemma 7.1): $E_k^2 \leq C_1\, \mathcal{E}_k^2(U_k, R_k)$, where $R_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$ is the refined set.
4. **Interior node property** (Rem. 5.4): $C_2\, \mathcal{E}_k^2(U_k, T) \leq |||U_{k+1} - U_k|||_{\omega_T}^2 + \mathrm{osc}_k^2(U_k, \omega_T)$, provided REFINE adds an interior node to each marked T.
5. **Pythagoras orthogonality:**
$$e_k^2 = |||u - U_{k+1}|||_\Omega^2 + |||U_{k+1} - U_k|||_\Omega^2 = e_{k+1}^2 + E_k^2, \tag{24}$$
because $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$ for $U_{k+1} - U_k \in \mathbb{V}_{k+1}$.

Properties 1, 2, 4 come from Lecture 5 and property 5 is Galerkin orthogonality in the energy inner product; property 3 is proved next, followed by the estimator reduction that the general contraction proof requires.

7.3 Localized upper bound

The energy distance between two consecutive Galerkin solutions is controlled by the estimator *restricted to the refined elements*: the discrete solution changes only where the mesh changes.

Lemma 7.1 (localized upper bound)**.** *Let $\mathcal{T}_{k+1} \geq \mathcal{T}_k$ with refined set $R_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$, and let $U_k \in \mathbb{V}_k$, $U_{k+1} \in \mathbb{V}_{k+1}$ be the Galerkin solutions. Then*

$$\boxed{E_k = |||U_{k+1} - U_k|||_\Omega \;\leq\; \sqrt{C_1}\, \mathcal{E}_k(U_k, R_k), \qquad \mathcal{E}_k(U_k, R_k) = \Big(\sum_{T \in R_k} \mathcal{E}_k^2(U_k, T)\Big)^{1/2},}$$

with C_1 the reliability constant of Theorem 5.1.

Proof. Set $V = U_{k+1} - U_k \in \mathbb{V}_{k+1}$ and let $I_k V \in \mathbb{V}_k$ be the quasi-interpolant of V onto the coarse space. Galerkin orthogonality on $\mathcal{T}_{k+1}$ gives $\mathcal{B}[U_{k+1}, V] = \langle f, V\rangle$, and on $\mathcal{T}_k$ gives $\mathcal{B}[U_k, I_k V] = \langle f, I_k V\rangle$, so with the residual $\langle R_k, w\rangle = \langle f, w\rangle - \mathcal{B}[U_k, w]$,

$$|||V|||_\Omega^2 = \mathcal{B}[U_{k+1} - U_k, V] = \langle f, V\rangle - \mathcal{B}[U_k, V] = \langle R_k, V\rangle = \langle R_k, V - I_k V\rangle\,,$$

the last equality by $\langle R_k, I_k V\rangle = 0$. Now $I_k V$ reproduces V on every element not touched by refinement, so $V - I_k V$ is supported on the patch of R_k. Repeating the residual estimate of Theorem 5.1, but summing only over R_k,

$$\langle R_k, V - I_k V\rangle \;\le\; C\,\mathcal{E}_k(U_k, R_k)\,\|V\|_\Omega.$$

Dividing by $\|V\|_\Omega$ yields the claim (the constant is the reliability constant C_1). □

7.4 Estimator reduction

The second key property tracks the estimator across a refinement step. It combines two effects: newly created elements are *smaller*, which shrinks their indicators, while the change of discrete solution from U_k to U_{k+1} *perturbs* the indicators in a controlled, Lipschitz manner.

Lemma 7.2 (estimator reduction)**.** *Let $\mathcal{T}_{k+1} \ge \mathcal{T}_k$ be obtained by bisecting (at least once) every element of the marked set $\mathcal{M}_k$, and set $\lambda := 1 - 2^{-1/d} \in (0,1)$. There is a constant $C_0 > 0$, depending only on shape regularity and the data, such that for every $\delta > 0$,*

$$\boxed{\mathcal{E}^2_{k+1}(U_{k+1}) \;\le\; (1+\delta)\Big[\mathcal{E}^2_k(U_k) - \lambda\,\mathcal{E}^2_k(U_k,\mathcal{M}_k)\Big] + (1+\delta^{-1})\,C_0\,E_k^2.}$$

Proof. Step 1 (reduction for a fixed function). Fix $V \in V_k \subset V_{k+1}$. When an element T is bisected, each child $T' \subset T$ has $h_{T'} \le 2^{-1/d} h_T$ (bisection halves the measure and $h \sim |T|^{1/d}$ for shape-regular elements). Since V is polynomial on T, it stays polynomial on each child, and the newly created interior edges carry *no* flux jump; comparing indicators on the children with the indicator on T,

$$\sum_{T' \in \mathcal{T}_{k+1},\, T' \subset T} \mathcal{E}^2_{k+1}(V, T') \;\le\; 2^{-1/d}\,\mathcal{E}^2_k(V,T).$$

Unrefined elements keep their indicator. Summing over $\mathcal{T}_k$ and isolating the marked elements (all of which are refined),

$$\mathcal{E}^2_{k+1}(V) \;\le\; \mathcal{E}^2_k(V) - (1-2^{-1/d})\,\mathcal{E}^2_k(V,\mathcal{M}_k) = \mathcal{E}^2_k(V) - \lambda\,\mathcal{E}^2_k(V,\mathcal{M}_k). \tag{25}$$

Step 2 (Lipschitz dependence on the discrete argument). The indicator depends Lipschitz-continuously on its discrete argument: for $W = W_1 - W_2$ the interior-residual difference on each element is $\operatorname{div}(A\nabla W)$ and the jump difference is $[\![\, A\nabla W \cdot \boldsymbol{n} \,]\!]$, so the inverse estimates on the fine mesh

$$h_T\,\|\operatorname{div}(A\nabla W)\|_{L^2(T)} + h_T^{1/2}\,\|[\![\, A\nabla W \cdot \boldsymbol{n} \,]\!]\|_{L^2(\partial T)} \;\lesssim\; \|\nabla W\|_{L^2(\omega_T)}$$

give

$$\big|\mathcal{E}_{k+1}(W_1) - \mathcal{E}_{k+1}(W_2)\big| \;\le\; \sqrt{C_0}\,\|W_1 - W_2\|_\Omega. \tag{26}$$

Step 3 (combine). Apply (26) with $W_1 = U_{k+1}$, $W_2 = U_k$, then Young's inequality $(a+b)^2 \le (1+\delta)a^2 + (1+\delta^{-1})b^2$, and finally (25) with $V = U_k$:

$$\begin{aligned}\mathcal{E}^2_{k+1}(U_{k+1}) &\le \big(\mathcal{E}_{k+1}(U_k) + \sqrt{C_0}\,E_k\big)^2 \le (1+\delta)\,\mathcal{E}^2_{k+1}(U_k) + (1+\delta^{-1})\,C_0\,E_k^2 \\ &\le (1+\delta)\Big[\mathcal{E}^2_k(U_k) - \lambda\,\mathcal{E}^2_k(U_k,\mathcal{M}_k)\Big] + (1+\delta^{-1})\,C_0\,E_k^2.\end{aligned}$$ □

Remark 7.3. Estimator reduction is the engine of the *general* contraction proof: unlike the interior node property, it needs no assumption on oscillation and no interior node — only that marked elements are bisected. The price is that it controls the *estimator* rather than the error, so it must be coupled to the energy error through Pythagoras and reliability.

8 Contraction property

We now prove that the adaptive loop *contracts*: each step reduces a suitable error quantity by a fixed factor $\rho < 1$, so $U_k \to u$ at a linear rate. We give the argument twice — first in a transparent special case, then in full generality.

8.1 A simple case: piecewise constant data (no oscillation)

When the data are piecewise constant on the initial mesh — $f \in \mathcal{P}_0(\mathcal{T}_0)$ and $A \in \mathcal{P}_0(\mathcal{T}_0)$, so that $r_T = f|_T$ and the jumps are edgewise constant — the oscillation $\operatorname{osc}_k(U_k)$ *vanishes* on every refinement of $\mathcal{T}_0$. The estimator is then equivalent to the energy error with no correction, and one can contract e_k *directly*, using the interior node property to convert Dörfler marking into an energy decrement.

Theorem 8.1 (contraction, vanishing oscillation)**.** *Assume* $\operatorname{osc}_k(U_k) = 0$ *and that* $\mathcal{T}_{k+1}$ *satisfies the interior node property. With Dörfler marking* (23) *at parameter* θ*,*

$$e_{k+1} \le \alpha\, e_k, \qquad \alpha = \Big(1 - \theta^2 \frac{C_2}{C_1}\Big)^{1/2} < 1.$$

Proof. By (24), $e_{k+1}^2 = e_k^2 - E_k^2$, so it suffices to bound E_k^2 from below. Using the interior node property (4), Dörfler marking (23), and the upper bound (1),

$$E_k^2 \;\ge\; C_2\, \mathcal{E}_k^2(U_k, \mathcal{M}_k) \;\ge\; \theta^2 C_2\, \mathcal{E}_k^2 \;\ge\; \theta^2 \frac{C_2}{C_1}\, e_k^2.$$

Hence $e_{k+1}^2 \le \big(1 - \theta^2 C_2/C_1\big) e_k^2$. Since $C_2 \le C_1$ and $0 < \theta < 1$, the factor $\alpha^2 \in (0,1)$. □

Iterating gives *linear convergence*: $e_k \le \alpha^k e_0 \to 0$. Two features of this argument are restrictive: it assumes $\operatorname{osc}_k = 0$, and it relies on the interior node property (hence on Refine placing a node inside each marked element). The general case of §8.2 removes both assumptions.

Remark 8.2 (is the interior node property necessary?). Yes. For $A = I$, $f = 1$, $\Omega = (0,1)^2$ with the criss-cross initial mesh $\mathcal{T}_0$, a single bisection step that adds *no* interior node leaves the Galerkin solution unchanged: $U_1 = U_0 = \frac{1}{12}\phi_0$. Then $E_0 = \|U_1 - U_0\|_\Omega = 0$ and $e_0 = e_1$: no contraction occurs. Only the further refinement $\mathcal{T}_2$, which introduces interior nodes, yields $U_2 \ne U_1$. The interior node property is what guarantees that marked elements actually contribute to the energy decrement.

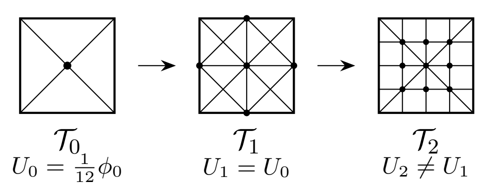


Figure 17: Counterexample illustrating necessity of the interior node property. $A = I$, $f = 1$, $\Omega = (0,1)^2$. Going from $\mathcal{T}_0$ to $\mathcal{T}_1$ adds midpoints on edges and diagonals but *no interior node*; the Galerkin solution is unchanged ($U_1 = U_0$) and the energy does not decrease. Only $\mathcal{T}_2$ introduces interior nodes, giving $U_2 \ne U_1$.

8.2 The general case

For general data the oscillation no longer vanishes and the interior node property may fail, so the energy error need not contract on its own. The counterexample of Figure 17 is the paradigm: there $U_{k+1} = U_k$, hence $e_{k+1} = e_k$ and the energy error *stalls*. Yet in exactly that situation the estimator reduction of Lemma 7.2 still bites: with $U_{k+1} = U_k$ (so $E_k = 0$) it reads, letting $\delta \to 0$,

$$\mathcal{E}_{k+1}^2(U_k) \;\le\; \mathcal{E}_k^2(U_k) - \lambda\, \mathcal{E}_k^2(U_k, \mathcal{M}_k),$$

a *strict* decrease of the estimator. Neither quantity contracts on its own — the energy error can stall while the estimator drops, and the estimator alone is not even monotone because it depends on the changing discrete solution — but a weighted combination of the two, the *quasi error*

$$\Delta_k := e_k^2 + \gamma\, \mathcal{E}_k^2 \qquad (\gamma > 0 \text{ to be chosen}),$$

does contract. This interplay between a *continuous* quantity (the energy error) and a *discrete* one (the estimator), neither of which suffices alone, is a hallmark of the convergence theory of Cascón, Kreuzer, Nochetto and Siebert.

Theorem 8.3 (contraction of the quasi error)**.** *Let $0 < \theta \le 1$ and suppose each step uses Dörfler marking (23) followed by* Refine *bisecting every marked element. Then there are constants $\gamma > 0$ and $\alpha \in (0,1)$, depending only on shape regularity, the data, and θ, such that the quasi error contracts:*

$$\boxed{\Delta_{k+1} = e_{k+1}^2 + \gamma\, \mathcal{E}_{k+1}^2 \;\le\; \alpha^2 \left(e_k^2 + \gamma\, \mathcal{E}_k^2\right) = \alpha^2\, \Delta_k \qquad \forall\, k \ge 0.}$$

In particular $e_k \le \Delta_k^{1/2} \le \alpha^k \Delta_0^{1/2}$, so AFEM converges linearly for any *right-hand side and coefficient — with or without oscillation, and without the interior node property.*

Proof. Set $E_k = \|U_{k+1} - U_k\|_\Omega$; we proceed in four steps.

Step 1 (assemble). Pythagoras (24) gives $e_{k+1}^2 = e_k^2 - E_k^2$, and estimator reduction (Lemma 7.2) gives, for any $\delta > 0$,

$$\mathcal{E}_{k+1}^2 \;\le\; (1+\delta)\big(\mathcal{E}_k^2 - \lambda\, \mathcal{E}_k^2(U_k, \mathcal{M}_k)\big) + (1+\delta^{-1}) C_0\, E_k^2.$$

Multiplying the second inequality by γ and adding the first,

$$\Delta_{k+1} \;\le\; e_k^2 + \big(\gamma(1+\delta^{-1})C_0 - 1\big)E_k^2 + \gamma(1+\delta)\big(\mathcal{E}_k^2 - \lambda\, \mathcal{E}_k^2(U_k, \mathcal{M}_k)\big).$$

Step 2 (choose δ and γ). Fix $\delta > 0$ by

$$(1+\delta)(1-\lambda\theta^2) = 1 - \tfrac{1}{2}\lambda\theta^2, \qquad \text{i.e.} \qquad \delta = \frac{\lambda\theta^2/2}{1-\lambda\theta^2} > 0,$$

and then $\gamma > 0$ by $\gamma(1+\delta^{-1})C_0 = 1$, which cancels the E_k^2 term exactly:

$$\Delta_{k+1} \;\le\; e_k^2 + \gamma(1+\delta)\big(\mathcal{E}_k^2 - \lambda\, \mathcal{E}_k^2(U_k, \mathcal{M}_k)\big).$$

Step 3 (Dörfler). Marking (23) gives $\mathcal{E}_k^2(U_k, \mathcal{M}_k) \ge \theta^2 \mathcal{E}_k^2$, so using the choice of δ,

$$\Delta_{k+1} \;\le\; e_k^2 + \gamma(1+\delta)(1-\lambda\theta^2)\, \mathcal{E}_k^2 = e_k^2 + \gamma\big(1 - \tfrac{1}{2}\lambda\theta^2\big)\mathcal{E}_k^2 = e_k^2 - \tfrac{1}{4}\gamma\lambda\theta^2\, \mathcal{E}_k^2 + \gamma\big(1 - \tfrac{1}{4}\lambda\theta^2\big)\mathcal{E}_k^2.$$

Step 4 (reliability closes the loop). The upper bound $e_k^2 \le C_1 \mathcal{E}_k^2$ (property 1) gives $\mathcal{E}_k^2 \ge e_k^2/C_1$, so $-\tfrac{1}{4}\gamma\lambda\theta^2\, \mathcal{E}_k^2 \le -\dfrac{\gamma\lambda\theta^2}{4C_1}\, e_k^2$, whence

$$\Delta_{k+1} \;\le\; \Big(1 - \frac{\gamma\lambda\theta^2}{4C_1}\Big)e_k^2 + \Big(1 - \frac{\lambda\theta^2}{4}\Big)\gamma\, \mathcal{E}_k^2 \;\le\; \alpha^2\, \Delta_k, \qquad \alpha^2 := \max\Big\{1 - \frac{\gamma\lambda\theta^2}{4C_1},\, 1 - \frac{\lambda\theta^2}{4}\Big\} < 1. \quad \square$$

Remark 8.4 (which ingredients each proof uses). The simple case contracts e_k itself but needs the interior node property and $\mathrm{osc}_k = 0$; the general case contracts the quasi error Δ_k and needs neither. The general proof rests on exactly four ingredients: Dörfler marking, the Pythagoras identity (24), the a posteriori *upper* bound (reliability), and the estimator reduction of Lemma 7.2. Notably it does *not* use the lower bound (efficiency): that property re-enters only in the rate analysis of Lecture 9.

Remark 8.5 (quasi error vs. total error; single marking). Reliability and efficiency together give $\mathcal{E}_k^2 \simeq e_k^2 + \mathrm{osc}_k^2$, so the quasi error is equivalent to the *total error* $e_k^2 + \mathrm{osc}_k^2$ — the quantity whose decay governs the optimal *rates* of Lecture 9. Thus efficiency, unused in the contraction itself, is exactly what ties the contracted quantity to the approximation classes. Note also that MARK is driven by the estimator *alone*: a single Dörfler step suffices. Earlier analyses marked separately for the estimator and for oscillation, which can degrade the rate; the quasi-error argument avoids this and, as Lecture 9 shows, still yields quasi-optimal convergence.

9 Convergence rates

Convergence (Lecture 8) says the error tends to zero; *optimality* says it does so at the best possible rate for the given solution — matching the rate of the best mesh with the same number of degrees of freedom. As with the contraction property, we proceed in two stages: first piecewise-constant data, where oscillation vanishes and the energy error alone measures the rate; then general data, where the natural quantity is the *total error.*

9.1 A simple case: piecewise constant data (no oscillation)

Throughout this subsection f and A are piecewise constant on $\mathcal{T}_0$, so $\mathrm{osc}_k \equiv 0$ on every refinement and the estimator is two-sided equivalent to the energy error, $C_2\,\mathcal{E}_k^2 \le e_k^2 \le C_1\,\mathcal{E}_k^2$. The achievable rate is then measured purely by the energy error.

Approximation classes. Let Π_N be the set of *all* conforming refinements of $\mathcal{T}_0$ (by bisection) with at most N elements more than $\mathcal{T}_0$, and define the best N-element error

$$\sigma_N(u) = \inf_{\mathcal{T}\in\Pi_N}\ \inf_{V\in V(\mathcal{T})} \|u - V\|_\Omega.$$

We say $u \in \mathbb{A}_s$ if

$$|u|_s := \sup_{N\in\mathbb{N}} N^s\,\sigma_N(u) < \infty, \qquad \text{i.e.} \qquad \sigma_N(u) \le |u|_s\, N^{-s}.$$

Thus $\mathbb{A}_s$ collects the functions approximable at rate N^{-s} by *some* adaptive mesh.

Remark 9.1. For $d = 2$, $-\Delta u = f$ on a polygon (including reentrant corners), one has $u \in \mathbb{A}_{1/2}$ with $|u|_{1/2} \lesssim |u|_{W_p^2}$, the same $N^{-1/2}$ that graded meshes deliver (Lecture 4). The point of this lecture is that AFEM *realizes* this rate *without* knowing the singularity in advance.

Exercise 9.2. Show $u \in \mathbb{A}_s$ iff for every $\varepsilon > 0$ there is a conforming refinement $\mathcal{T}_\varepsilon \ge \mathcal{T}_0$ and $V_\varepsilon \in V(\mathcal{T}_\varepsilon)$ with $\|u - V_\varepsilon\|_\Omega \le \varepsilon$ and $\#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |u|_s^{1/s}\,\varepsilon^{-1/s}$.

Optimal marking. The next lemma is the bridge between energy reduction and the bulk-marking criterion: any mesh that reduces the error enough must mark a bulk fraction of the estimator.

Lemma 9.3 (optimal marking)**.** *Let $\theta < \theta_* = \sqrt{C_2/C_1}$, so $\mu := 1 - \theta^2/\theta_*^2 > 0$. If a refinement $\mathcal{T}_* \geq \mathcal{T}_k$ with Galerkin solution U_* satisfies $|||u - U_*|||_\Omega^2 \leq \mu\, |||u - U_k|||_\Omega^2$, then the refined set $R = \mathcal{T}_k \setminus \mathcal{T}_*$ satisfies the Dörfler property $\mathcal{E}_k(U_k, R) \geq \theta\, \mathcal{E}_k(U_k, \mathcal{T}_k)$.*

Proof. Using lower bound, the energy reduction hypothesis, Pythagoras (24), and the localized upper bound (assuming $\mathrm{osc}_k = 0$ for clarity),

$$\begin{aligned}(1-\mu) C_2\, \mathcal{E}_k^2 \leq (1-\mu)\, e_k^2 = e_k^2 - \mu\, e_k^2 &\leq |||u - U_k|||_\Omega^2 - |||u - U_*|||_\Omega^2 \\ &= |||U_k - U_*|||_\Omega^2 \leq C_1\, \mathcal{E}_k^2(U_k, R).\end{aligned}$$

Thus $(1-\mu)\frac{C_2}{C_1}\mathcal{E}_k^2 \leq \mathcal{E}_k^2(U_k, R)$, i.e. $\mathcal{E}_k(U_k, R) \geq \theta\, \mathcal{E}_k$ with $\theta^2 = (1-\mu)C_2/C_1$. ☐

Cardinality of the marked set.

Lemma 9.4 (cardinality)**.** *Assume $\theta < \theta_*$ and that* Mark *selects a set of minimal cardinality. If $u \in \mathbb{A}_s$, then for all k,*

$$\#\mathcal{M}_k \;\lesssim\; |u|_s^{1/s}\, |||u - U_k|||_\Omega^{-1/s}.$$

Sketch. Apply the Exercise with $\varepsilon^2 = \mu\, e_k^2$ to get a mesh $\mathcal{T}_\varepsilon$ achieving $|||u - U_\varepsilon|||_\Omega \leq \varepsilon$ with $\#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |u|_s^{1/s}\, \varepsilon^{-1/s}$. Overlay $\mathcal{T}_* = \mathcal{T}_k \oplus \mathcal{T}_\varepsilon$ (smallest common refinement); then $|||u - U_*|||_\Omega \leq \varepsilon$, so $\mathcal{T}_*$ meets the hypothesis of Lemma 9.3 and its refined set is Dörfler. Minimal cardinality of $\mathcal{M}_k$ and $\#\mathcal{T}_* - \#\mathcal{T}_k \leq \#\mathcal{T}_\varepsilon - \#\mathcal{T}_0$ give the bound. ☐

Quasi-optimality.

Theorem 9.5 (quasi-optimal rate, no oscillation)**.** *Assume* $\mathrm{osc} \equiv 0$ *(piecewise-constant data), $\theta < \theta_* = \sqrt{C_2/C_1}$, the interior node property, and minimal-cardinality Dörfler marking. If $u \in \mathbb{A}_s$ then AFEM produces*

$$|||u - U_k|||_\Omega \;\lesssim\; |u|_s \,\big(\#\mathcal{T}_k - \#\mathcal{T}_0\big)^{-s} \qquad \forall\, k \geq 1.$$

That is, AFEM converges with the best possible rate N^{-s} for the class $\mathbb{A}_s$.

Sketch. By the complexity bound (Theorem 4.5) and Lemma 9.4,

$$\#\mathcal{T}_k - \#\mathcal{T}_0 \;\lesssim\; \sum_{j=0}^{k-1} \#\mathcal{M}_j \;\lesssim\; |u|_s^{1/s} \sum_{j=0}^{k-1} e_j^{-1/s}.$$

Contraction (Theorem 8.1, valid here since $\mathrm{osc} \equiv 0$) gives geometric decay of the error, $e_k \leq \alpha^{k-j} e_j$ for $j \leq k$, i.e. $e_j \geq \alpha^{-(k-j)} e_k$; hence $e_j^{-1/s} \leq \alpha^{(k-j)/s} e_k^{-1/s}$, and the geometric series $\sum_j (\alpha^{1/s})^{k-j}$ converges since $\alpha < 1$. Therefore $\#\mathcal{T}_k - \#\mathcal{T}_0 \lesssim |u|_s^{1/s}\, e_k^{-1/s}$, which rearranges to the claim. ☐

9.2 The general case: the total error

For general f, A the oscillation no longer vanishes, and the energy error alone no longer determines the achievable rate. What AFEM actually reduces is the quasi error of Lecture 8, which by reliability and efficiency is equivalent to the *total error*

$$\mathcal{E}_{\mathcal{T}}^{\mathrm{tot}}(U)^2 \;:=\; |||u - U|||_\Omega^2 + \mathrm{osc}_{\mathcal{T}}^2(U) \;\simeq\; \mathcal{E}_{\mathcal{T}}^2(U) \;\simeq\; |||u - U|||_\Omega^2 + \gamma\, \mathcal{E}_{\mathcal{T}}^2(U).$$

The first equivalence is Theorems 5.1–5.5 ($C_2\mathcal{E}^2 \leq e^2 + \mathrm{osc}^2$ and $e^2 \leq C_1\mathcal{E}^2$); it is *here*, not in the contraction, that efficiency (the lower bound) is indispensable.

Approximation classes on the triple (u, f, A). Because oscillation depends on the data, the approximation class must be defined for the triple (u, f, A), not for u alone. With Π_N as before,

$$\sigma_N(u, f, A) = \inf_{\mathcal{T}\in\Pi_N} \inf_{V\in V(\mathcal{T})} \Big(\|u - V\|_\Omega^2 + \operatorname{osc}^2_{\mathcal{T}}(V) \Big)^{1/2},$$

and we say $(u, f, A) \in \mathbb{A}_s$ if $|u, f, A|_s := \sup_N N^s \sigma_N(u, f, A) < \infty$. For piecewise-constant A the oscillation reduces to *data oscillation* $\operatorname{osc}_{\mathcal{T}}(f) = \big\|h(f - \bar{f})\big\|_{L^2(\Omega)}$, and the class factorizes, $|u, f, A|_s \simeq |u|_s + |f|_{\mathbb{B}_s}$, recovering the energy-only class of §9.1 when f too is piecewise constant.

Quasi-optimality of the total error. A Céa-type lemma holds for the total error: on any admissible mesh the AFEM total error is comparable to the *best* total error,

$$\|u - U\|_\Omega^2 + \operatorname{osc}^2_{\mathcal{T}}(U) \;\lesssim\; \inf_{V\in V(\mathcal{T})} \Big(\|u - V\|_\Omega^2 + \operatorname{osc}^2_{\mathcal{T}}(V) \Big).$$

Hence the total error is the quantity AFEM quasi-minimizes, and $\mathbb{A}_s$ is the right notion of approximability for its decay.

Quasi-optimal rates.

Theorem 9.6 (quasi-optimal rate, general data)**.** *Let* Mark *use minimal-cardinality Dörfler marking with a threshold $\theta < \theta_* = \sqrt{C_2/\big(C_1(1 + C_1(1 + C_3))\big)}$, where $C_3 \propto \operatorname{osc}^2_{\mathcal{T}_0}(A)$ accounts for the oscillation. If $(u, f, A) \in \mathbb{A}_s$, then AFEM produces*

$$\|u - U_k\|_\Omega + \operatorname{osc}_k(U_k) \;\lesssim\; |u, f, A|_s \left(\#\mathcal{T}_k - \#\mathcal{T}_0\right)^{-s} \qquad \forall\, k \geq 1.$$

Sketch. The argument parallels §9.1 with the total error replacing the energy error; three ingredients change. *(i) Optimal marking:* if $\mathcal{T}_* \geq \mathcal{T}_k$ reduces the total error by a factor $\mu < 1$, its refined set is Dörfler for $\mathcal{E}_k$; the proof now carries an oscillation decomposition over the refined set $R = \mathcal{T}_k \backslash \mathcal{T}_*$ and its complement, which forces the stricter threshold θ_* above. *(ii) Cardinality:* overlaying $\mathcal{T}_k$ with a near-best mesh for the total error and using minimal cardinality gives $\#\mathcal{M}_k \lesssim |u, f, A|_s^{1/s} \big(\|u - U_k\|_\Omega^2 + \operatorname{osc}_k^2\big)^{-1/(2s)}$. *(iii) Contraction:* the quasi-error contraction of Theorem 8.3 (*not* the energy contraction, and needing *no* interior node property) supplies the geometric decay required to sum the cardinalities. Combined with the complexity bound (Theorem 4.5) this yields the stated rate. □

Remark 9.7 (what the general theory buys). It removes the two artificial hypotheses of §9.1 — vanishing oscillation and the interior node property — at the price of (a) a more stringent threshold θ_*, (b) measuring approximability of the triple (u, f, A) by the total error, and (c) driving the whole analysis by the quasi error. Remarkably, a *single* Dörfler marking on the estimator still suffices; no separate marking of oscillation is required.

Remark 9.8 (the complete picture). The three pillars combine cleanly: *reliability + efficiency* (Lecture 5) make the estimator a faithful proxy for the error; the *contraction property* (Lecture 8) gives convergence; and *optimal marking + complexity* (this lecture, with Lecture 4) upgrade convergence to the best-possible rate. Crucially, AFEM achieves the rate of the optimal graded mesh without any a priori knowledge of where the solution is singular — it discovers the right mesh automatically from the computed indicators.

References

[1] R. Verfürth, A note on polynomial approximation in Sobolev spaces, *ESAIM: Mathematical Modelling and Numerical Analysis*, 33(4):715–719, 1999.